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Medved'

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(45) **Date of Patent: Sep. 16, 2025**

(54) **FASTENING ARRANGEMENT AND SEAT**

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(73) Assignee: **Adient US LLC**, Plymouth, MI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 273 days.

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(30) **Foreign Application Priority Data**

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Dec. 10, 2021 (DE) 10 2021 214 160.9

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B60N 2/90 (2018.01)

(52) **U.S. Cl.**
CPC ***B60N 2/01583*** (2013.01); ***B60N 2/01516*** (2013.01); ***B60N 2/919*** (2018.02)

(58) **Field of Classification Search**
CPC B60N 2/01583; B60N 2/0881; B60N 2/08; B60N 2/0887; B60N 2/0893; B60N 2/0818; B60N 2/919; B60N 2/3097; B60N 2/30; B60N 2/0155; B60N 2/01508; B60N 2/01516; B60N 2002/952
USPC 296/63, 65.03, 65.15, 65.18; 297/440.14, 297/331, 335, 336, 440.22
See application file for complete search history.

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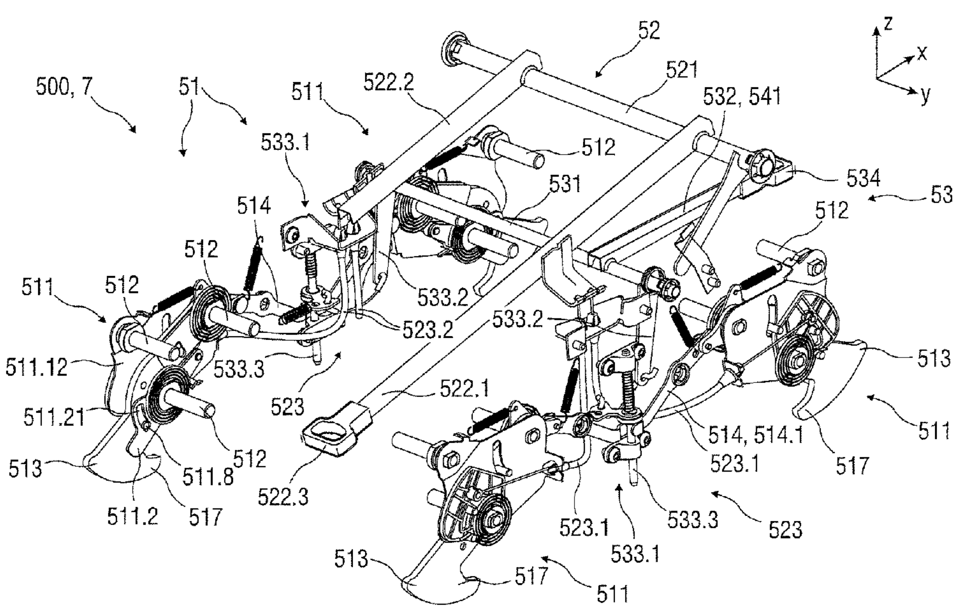
Primary Examiner — Amy R Weisberg
Assistant Examiner — Wenwei Zhuo
(74) *Attorney, Agent, or Firm* — Marshall & Melhorn, LLC

(57) **ABSTRACT**

A fastening arrangement of a seat on a rail arrangement may have a modular construction and may have one lock module for releasably arresting the fastening arrangement on the rail arrangement. The fastening arrangement may also have one lock unlocking module for unlocking the lock module. The fastening arrangement may also have one rail unlocking module.

9 Claims, 35 Drawing Sheets



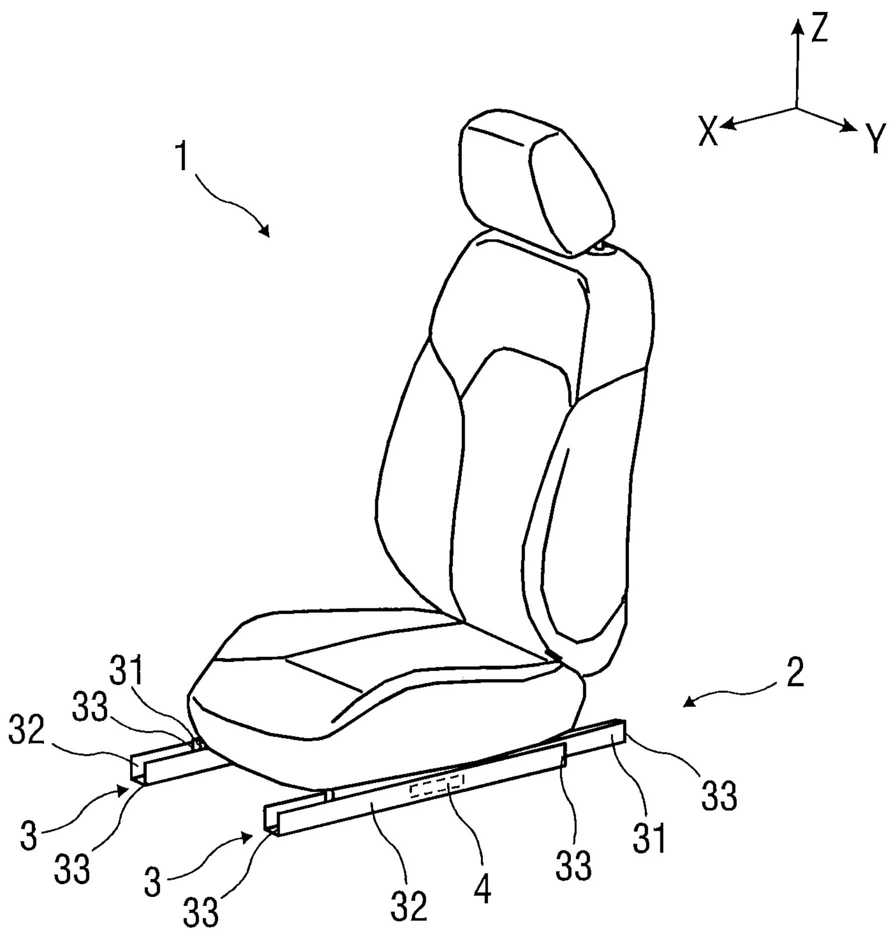
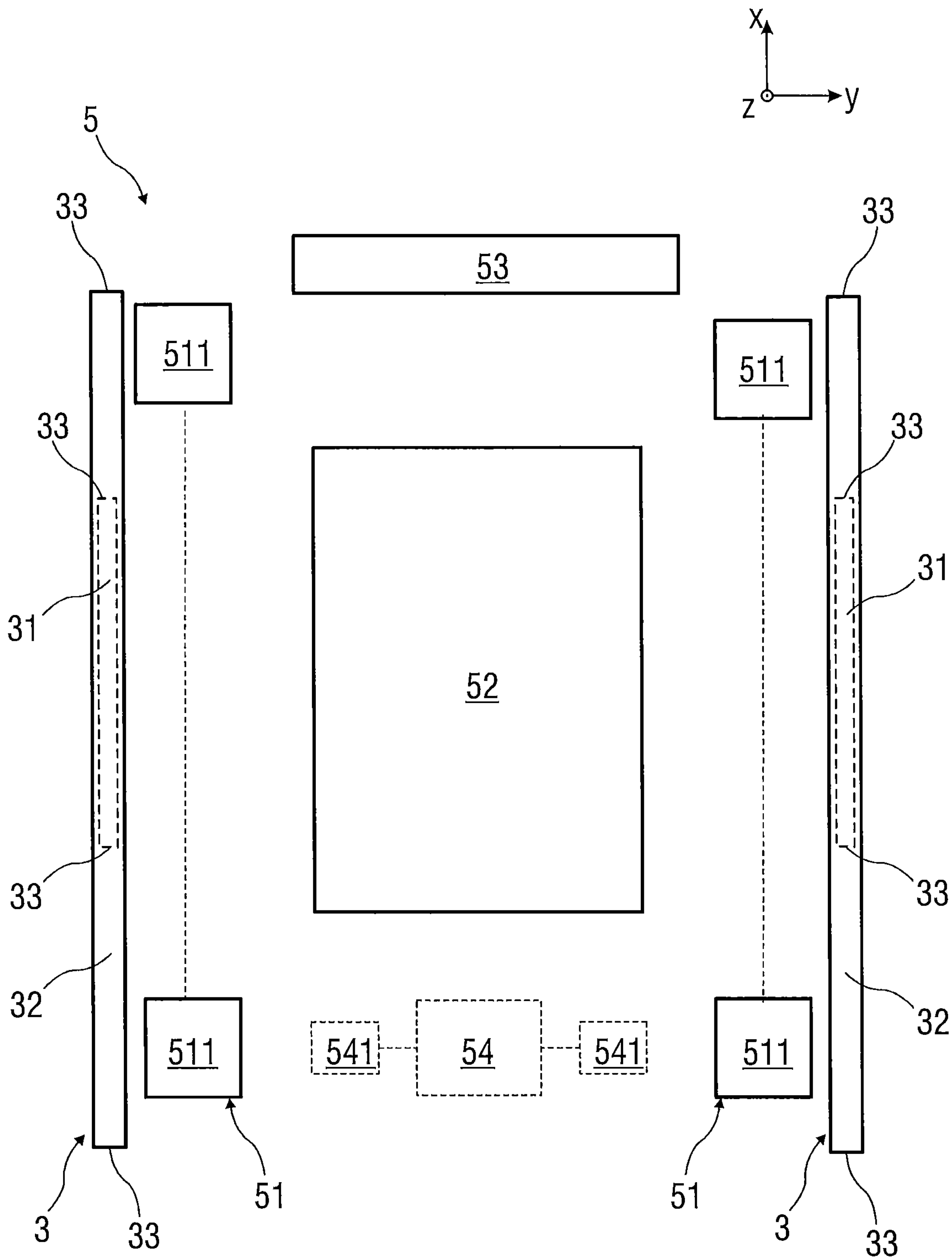


FIG 1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y
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33
53
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511
33
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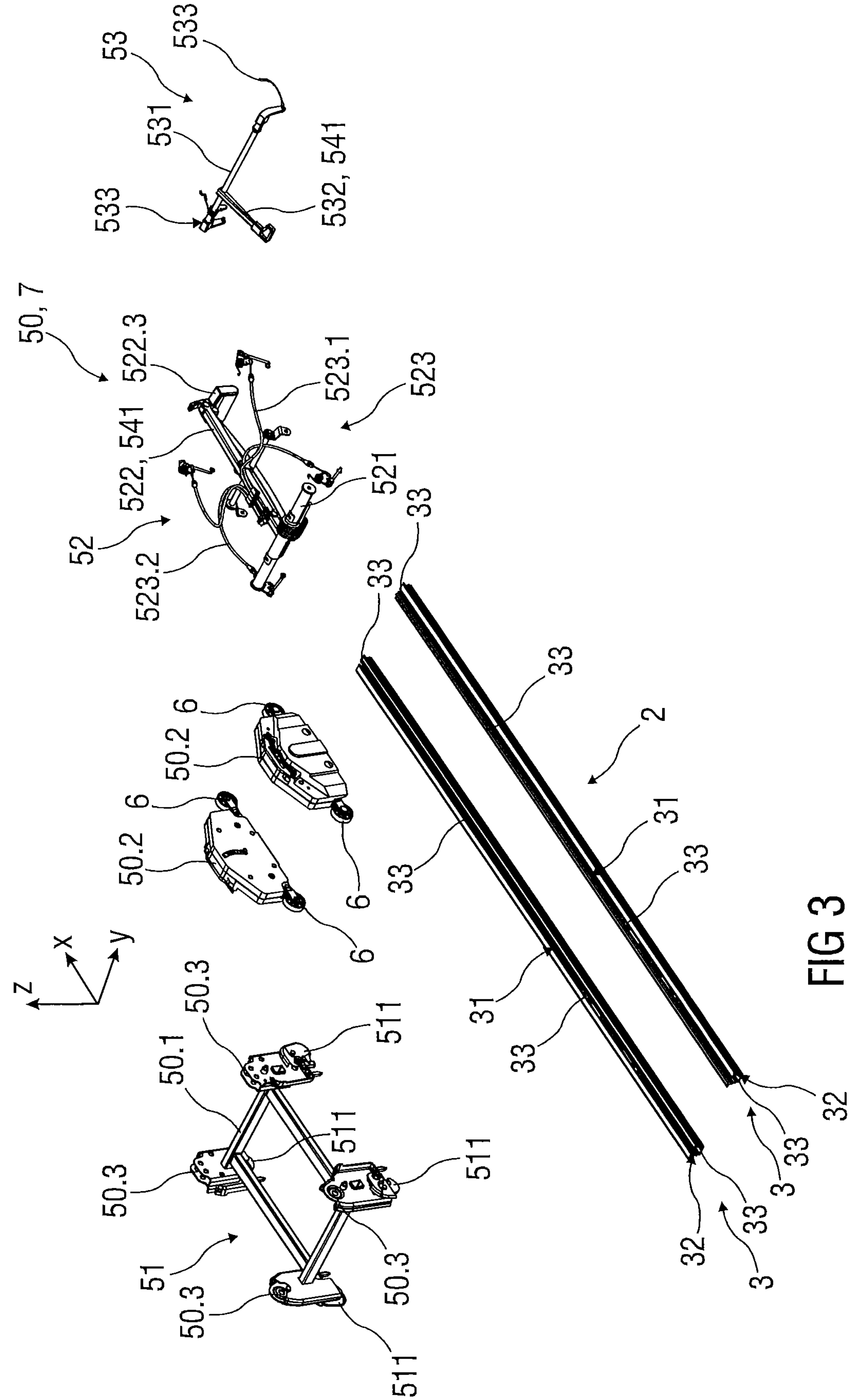
541
54
541
511
51
51
3
33
3
33

FIG 2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532, 541
50, 7
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522, 541
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523.2
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33
33
33
2
6
50.2
6
50.2
31
33
33
6
6
z
x
y
50.3
511
50.1
511
511
50.3
31
33
32
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3
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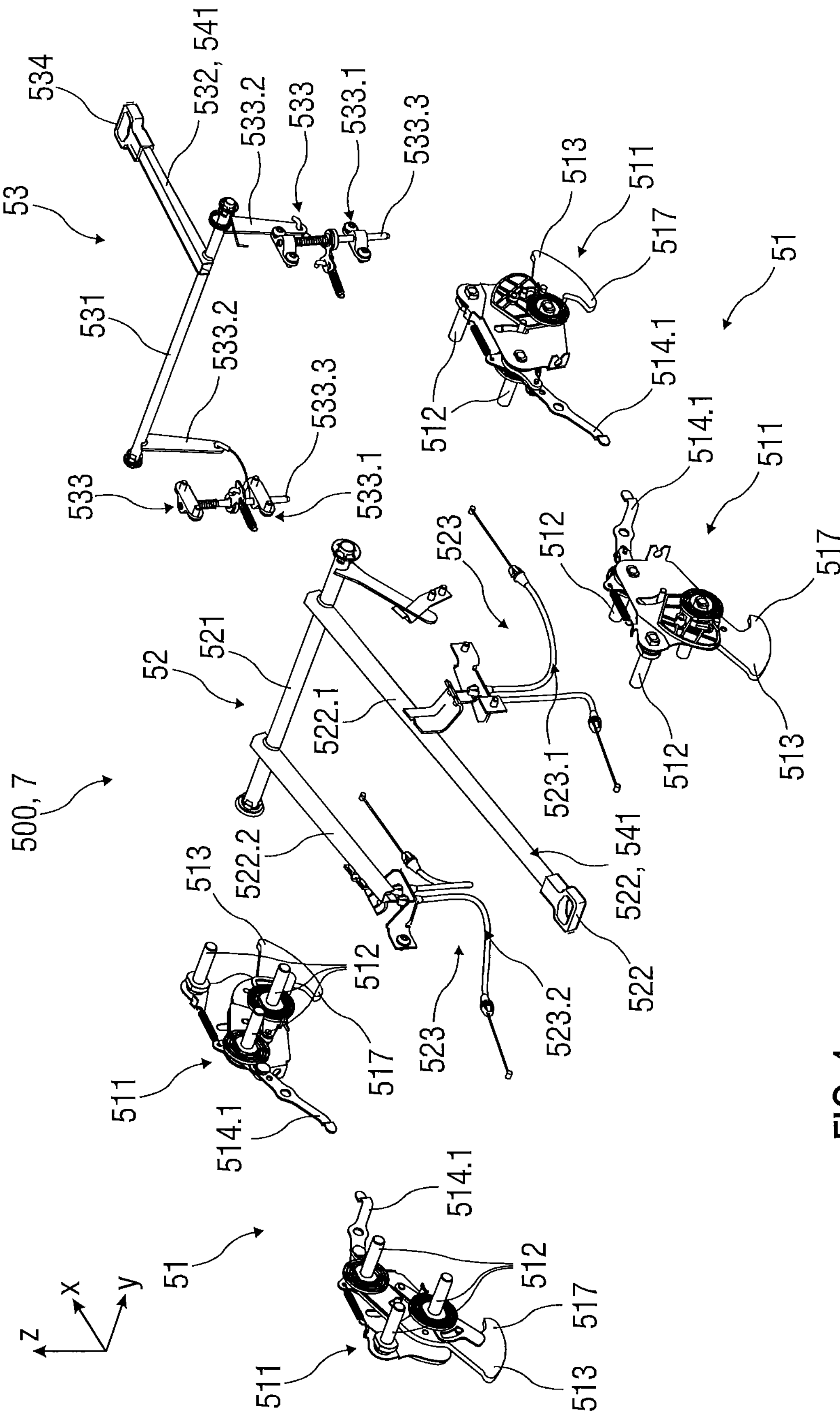
32
51
33
3
50.3
50.3
511

FIG 3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533.1
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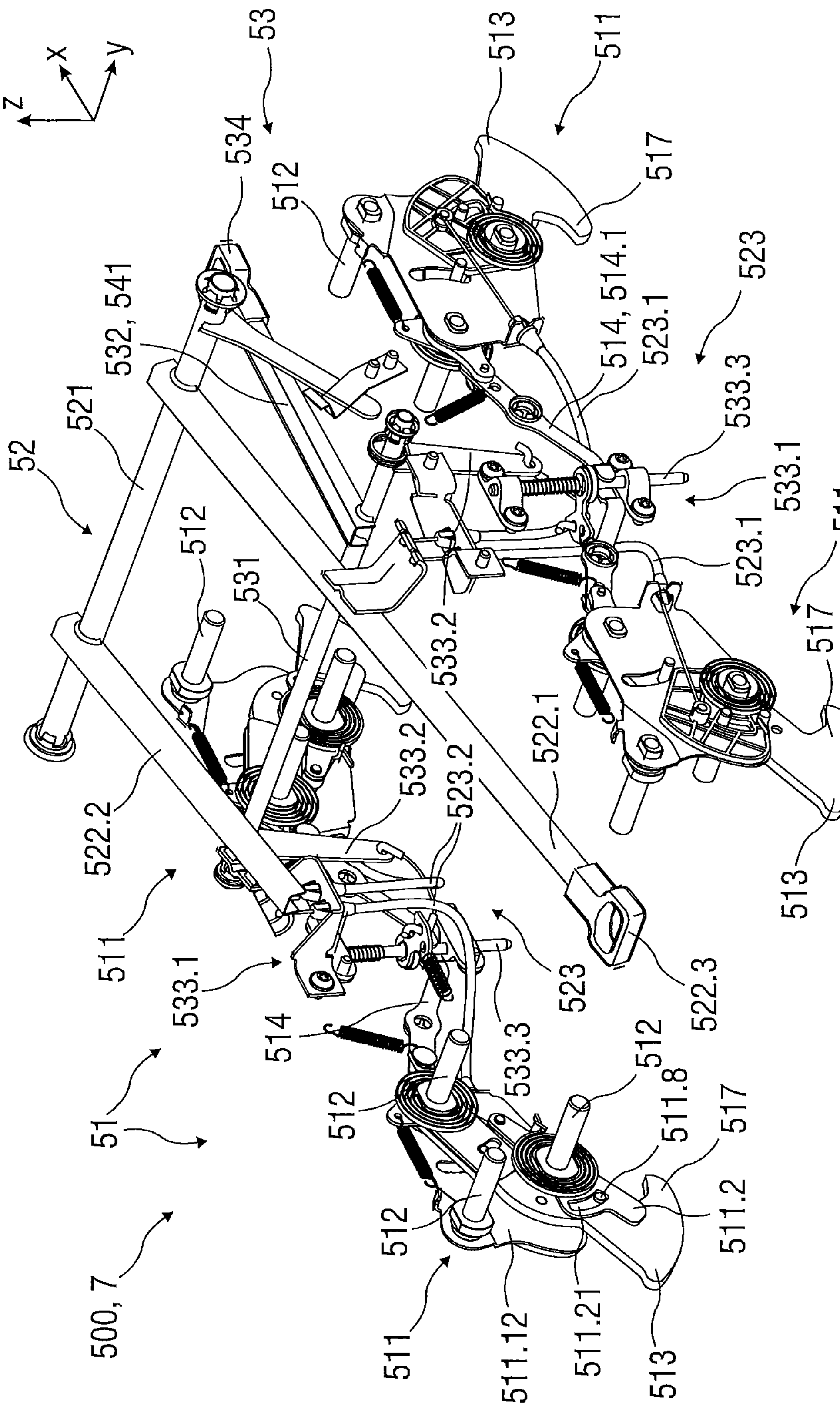
51
514.1
512
514.1
511
512
517
513
512
52
521
522.1
523
523.1
500, 7
522.2
513
512
517
511
514.1
523
523.2
522
522, 541
51
514.1
512
517
513
511
x
y
z

FIG 4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y
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52
523
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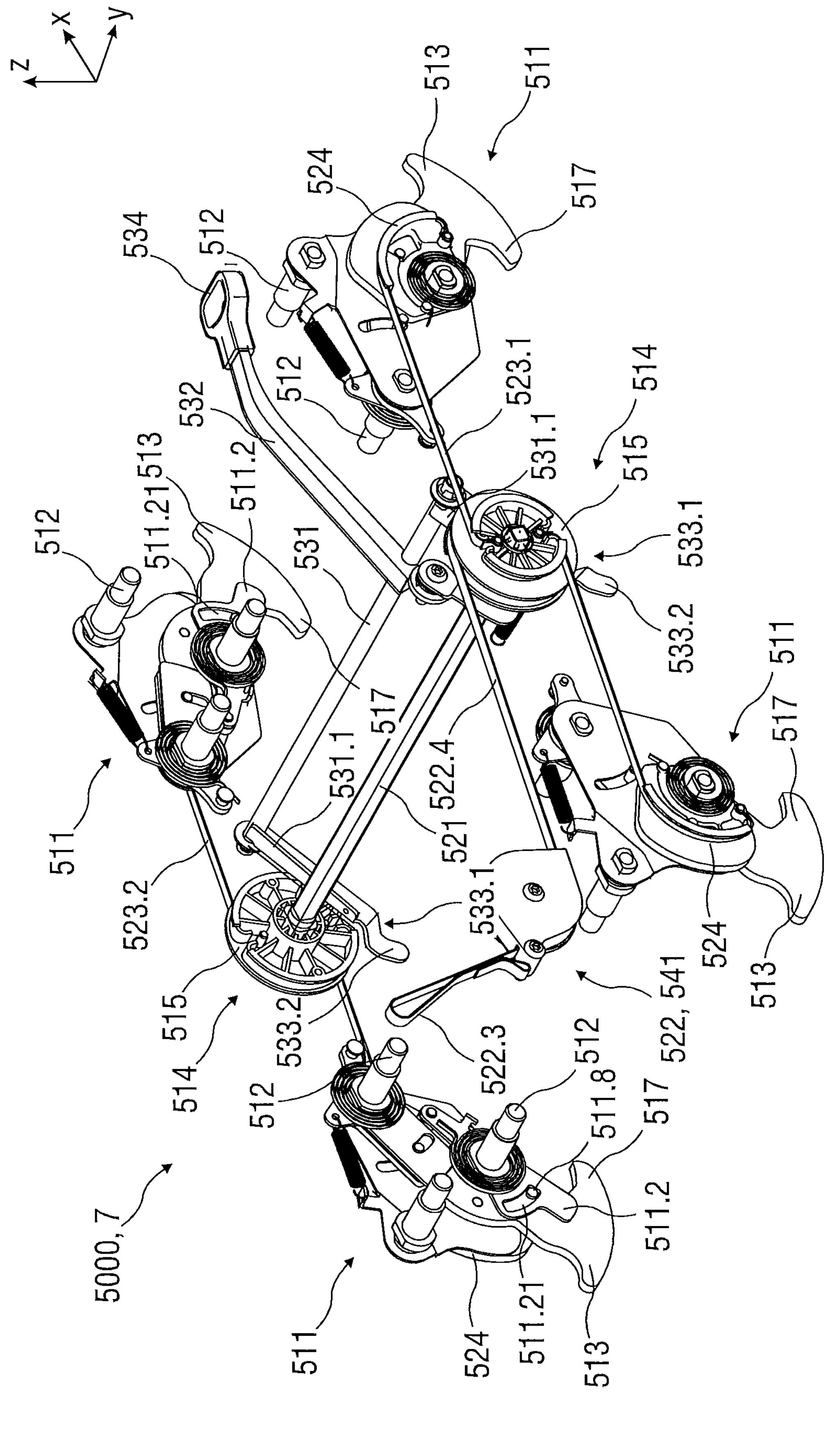
523.1
533.3
533.1
531
533.2
522.1
522.2
523.2
511
533.1
514
523
522.3
51
500, 7
511.12
511.21
511.8
511.2

FIG 5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y
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513
524
517
534
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523.1
514
532
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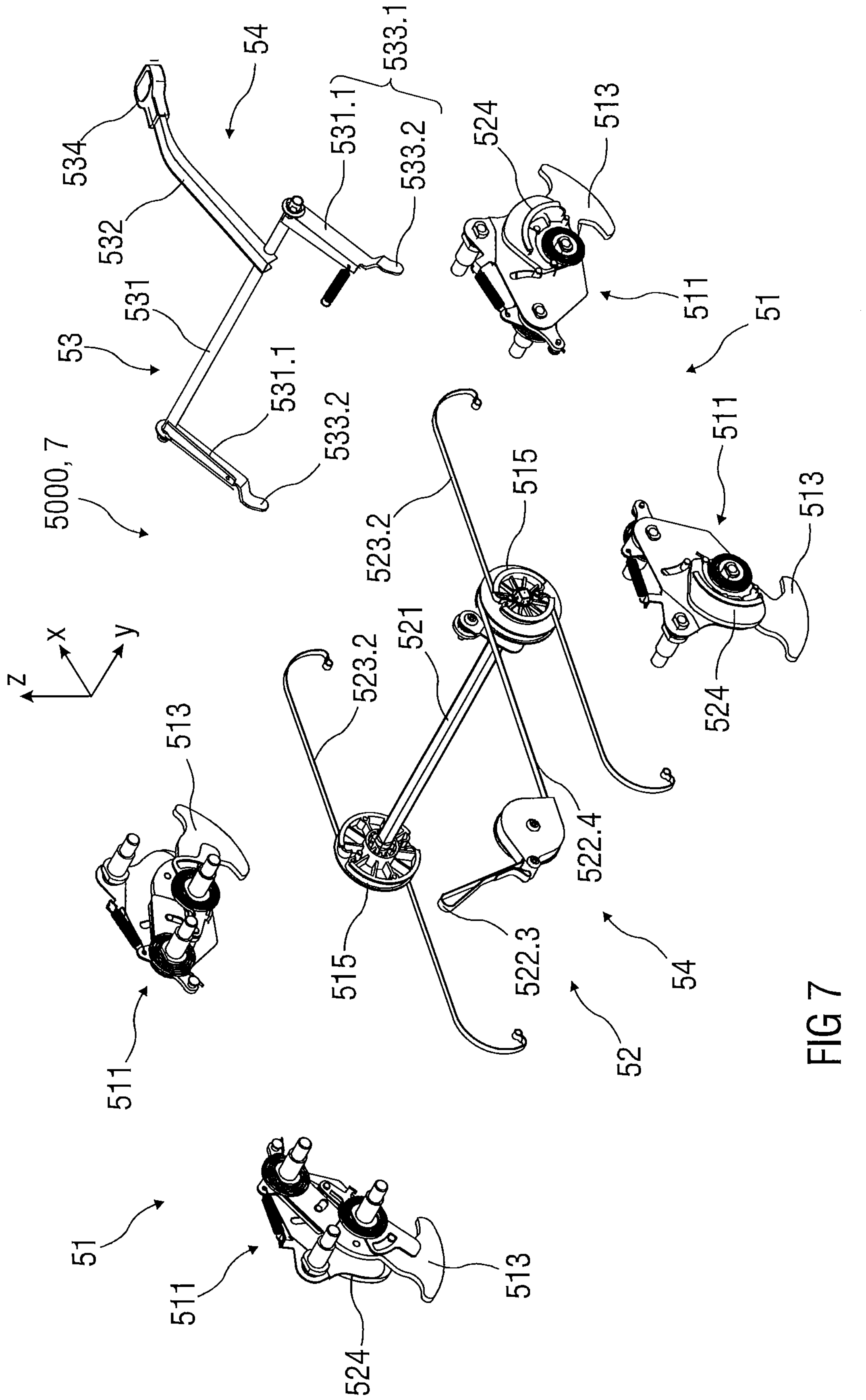
515
513
511.21
511.2
531
533.1
533.2
517
511
512
511
531.1
522.4
517
521
523.2
533.1
524
513
515
533.2
522, 541
514
522.3
512
512
511.8
517
5000, 7
511.2
511
524
511.21
513

FIG 6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533.2
532
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53
531.1
533.2
5000, 7
524
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511
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511
515
523.2
521
513
524
z
x
y
523.2
513
522.4
522.3
515
54
52
511
51
511
513
524

FIG 7

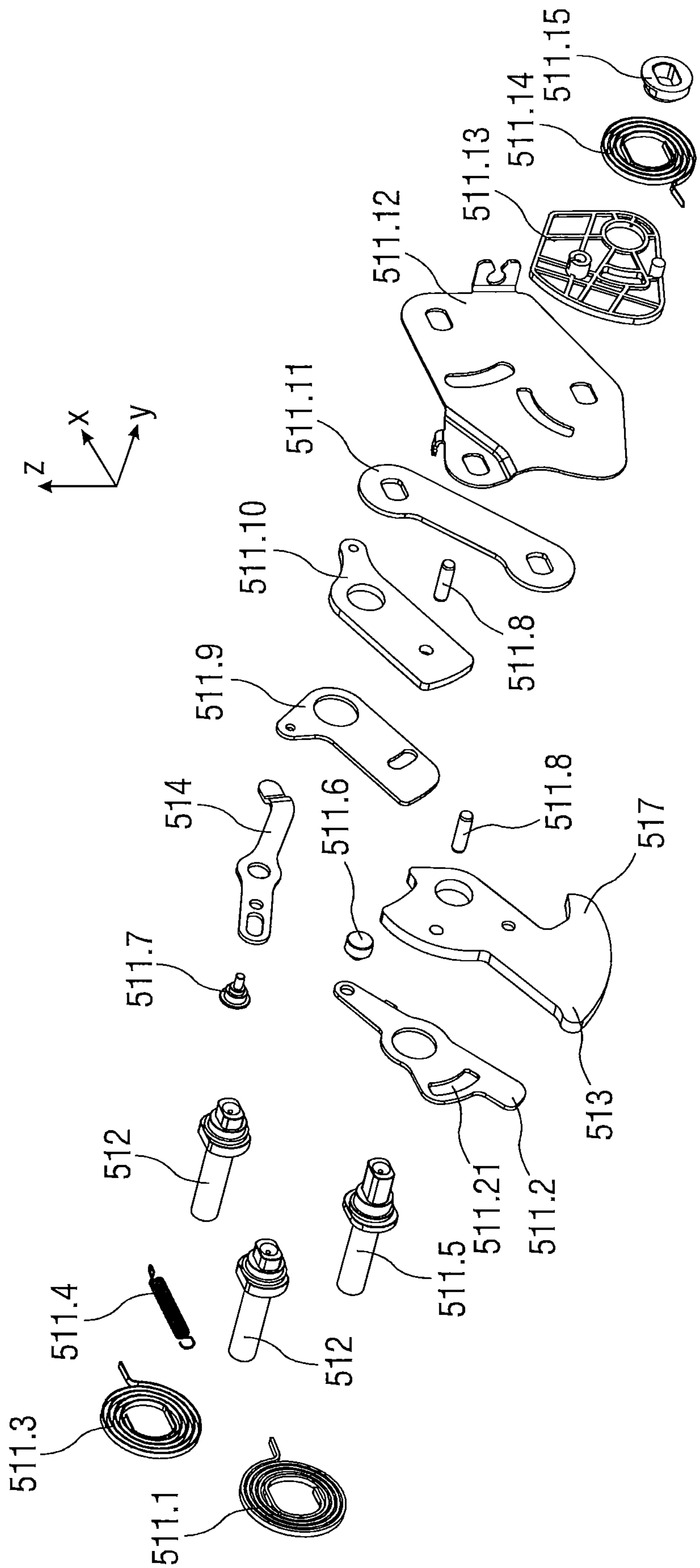

511.15
511.14
511.13
511.12
511.11
511.10
511.9
511.8
514
511.6
511.8
517
511.7
513
512
511.21
511.2
511.5
511.4
512
511.3
511.1
z
x
y

FIG 8

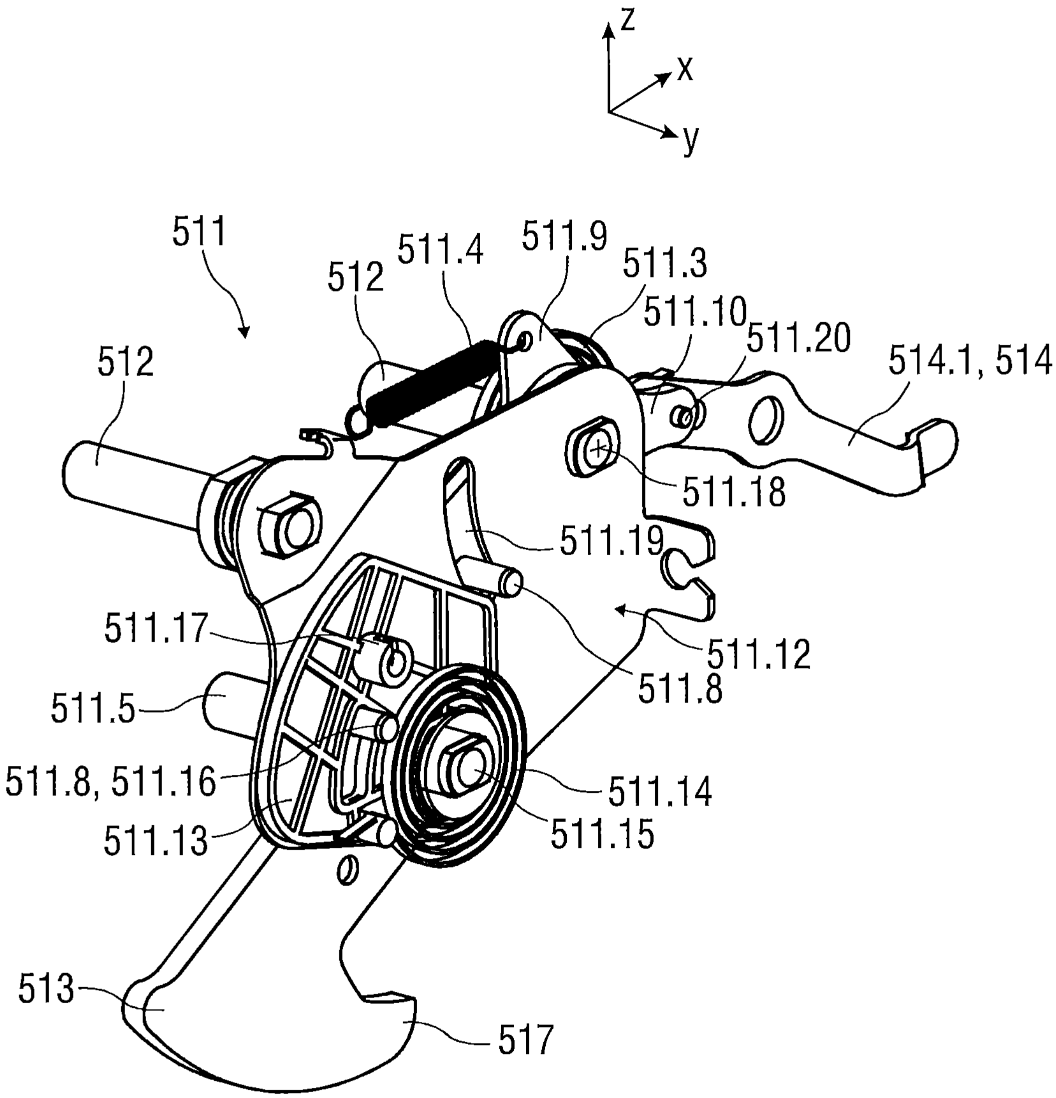

z
x
y
511
512
511.4
511.9
511.3
511.10
511.20
514.1, 514
512
511.18
511.19
511.17
511.12
511.8
511.5
511.8, 511.16
511.14
511.13
511.15
513
517

FIG 9

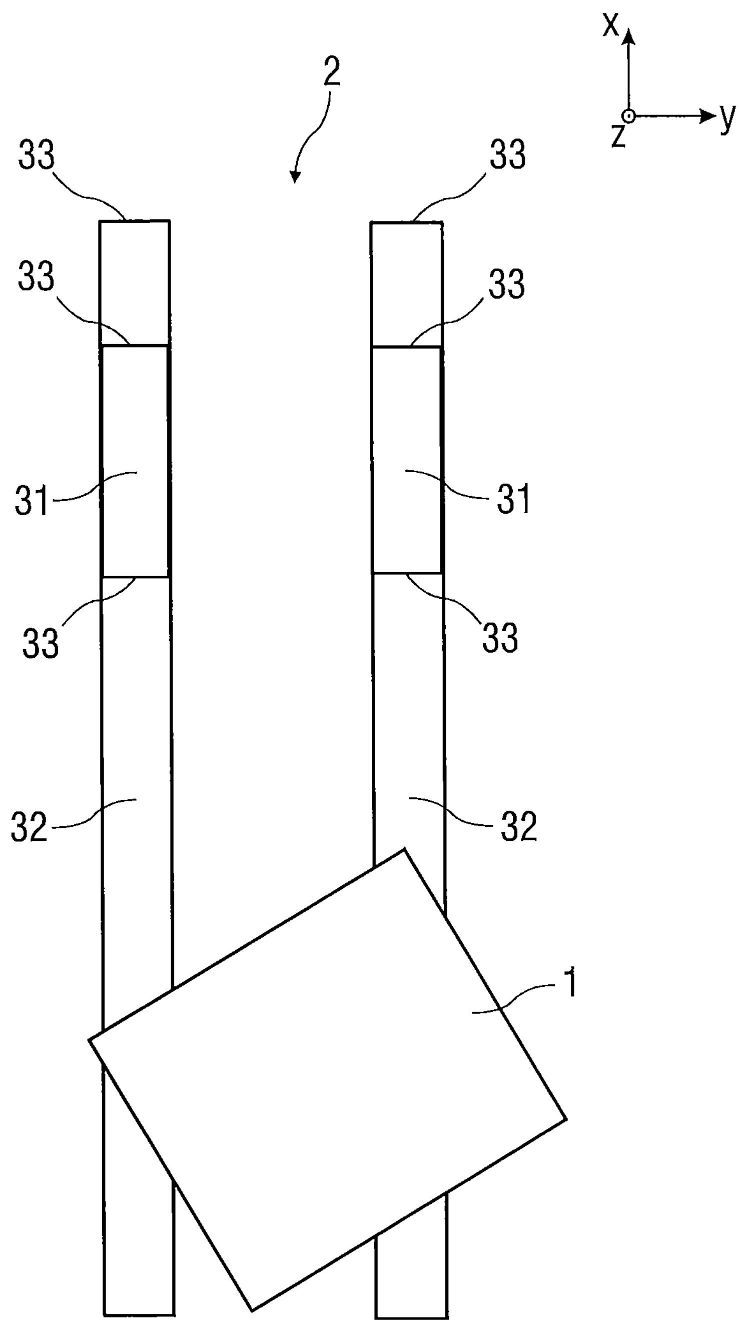


FIG 10

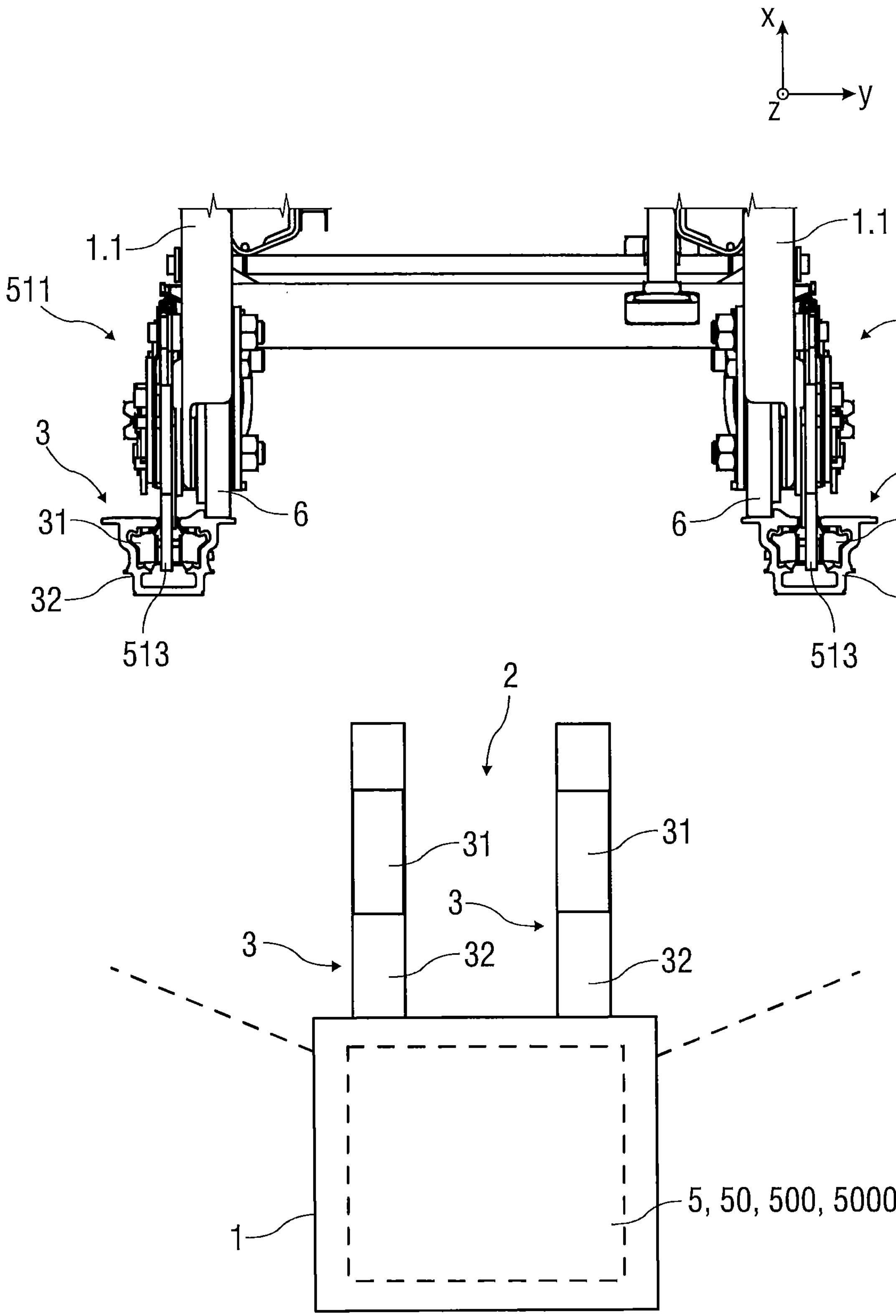

x
y
z
1.1
1.1
511
511
3
3
31
31
32
32
6
6
513
513
2
31
31
3
3
32
32
1
5, 50, 500, 5000

FIG 11

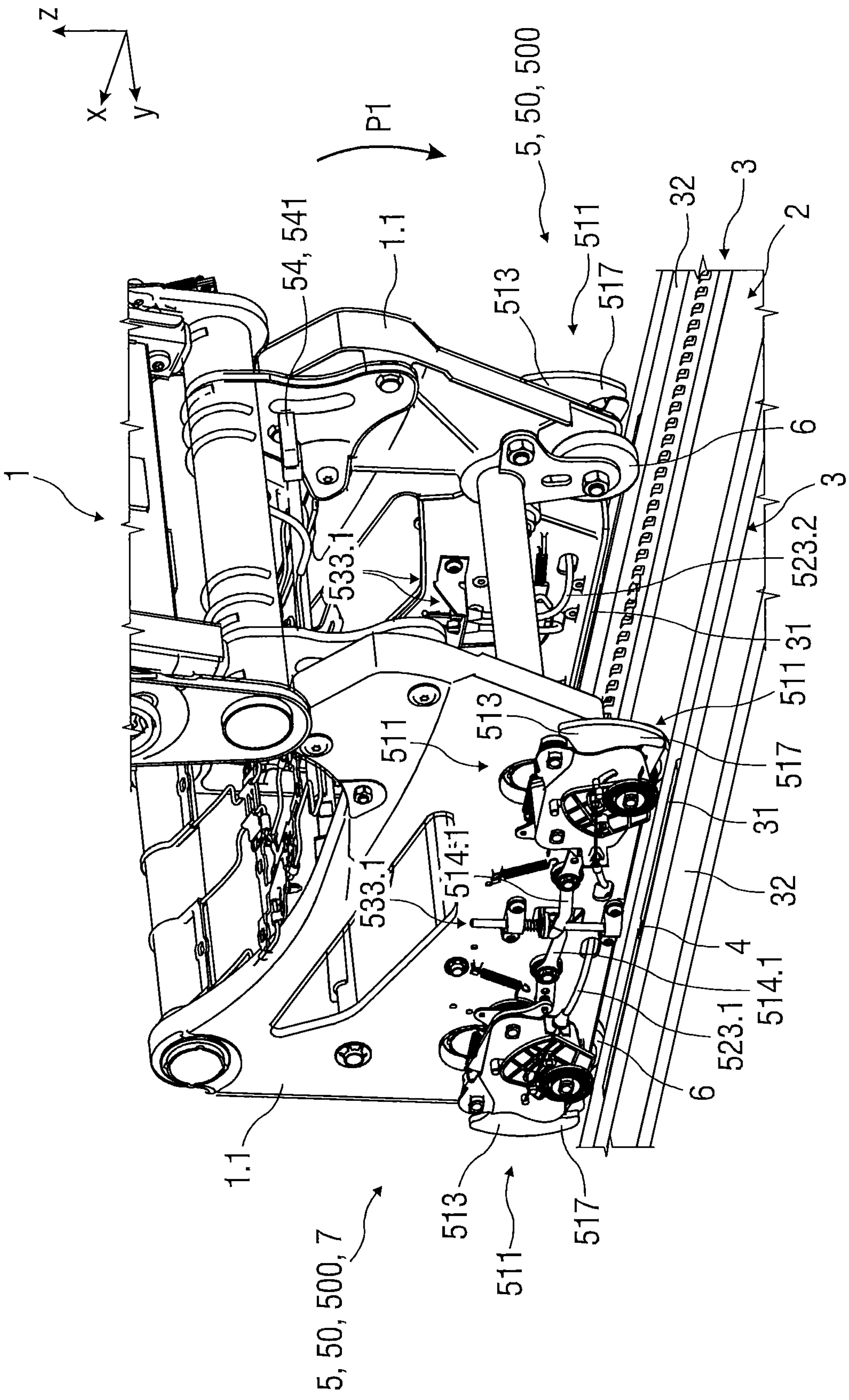

z
x
y
P1
5, 50, 500
1
54, 541
1.1
513
511
517
32
3
2
6
533.1
523.2
31
511
514.1
4
523.1
1.1
5, 50, 500, 7

FIG 12

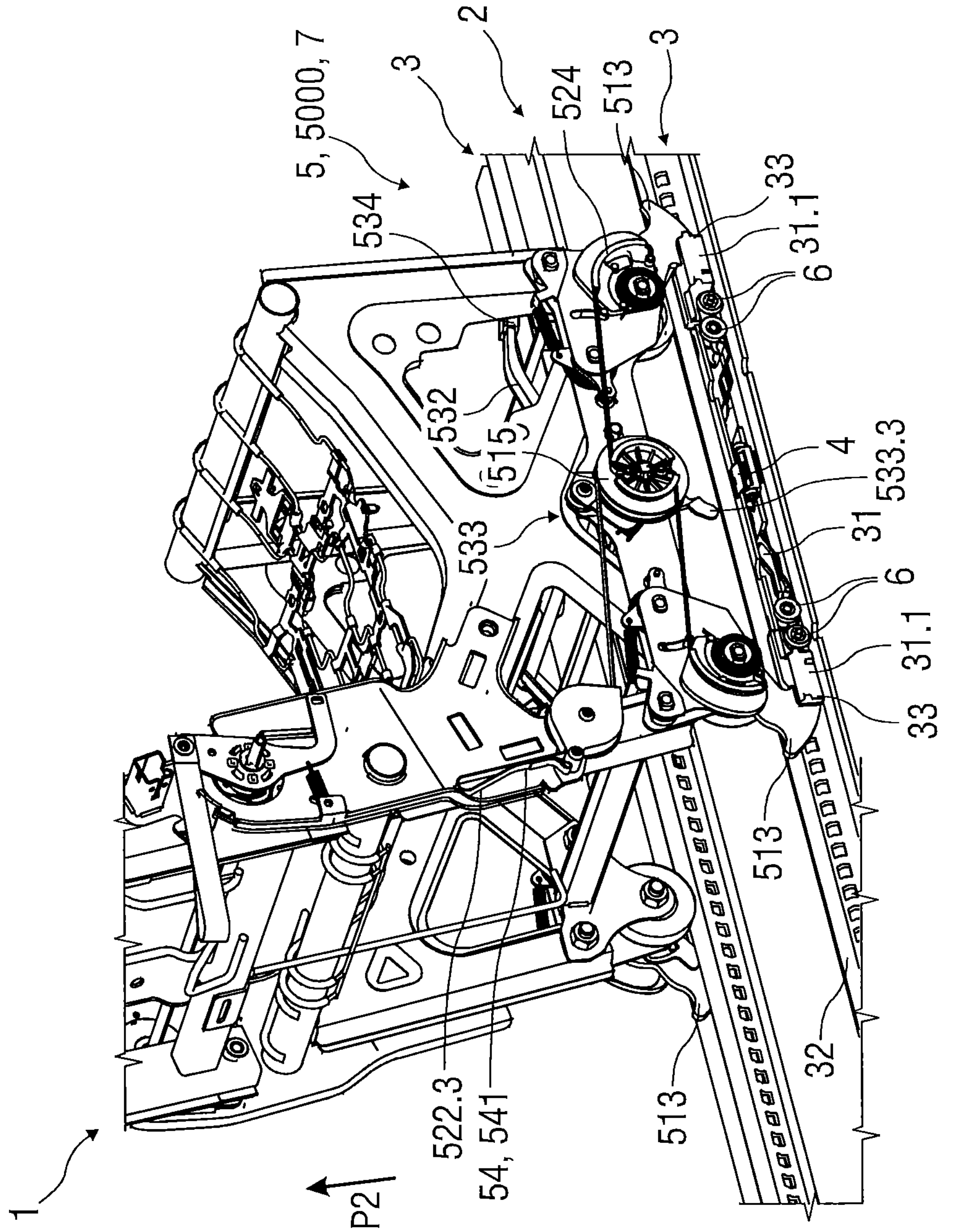

1
P2
5, 5000, 7
3
2
3
534
524
513
33
31.1
6
532
515
533
4
533.3
31
6
31.1
33
513
522.3
54, 541
513
32

FIG 13

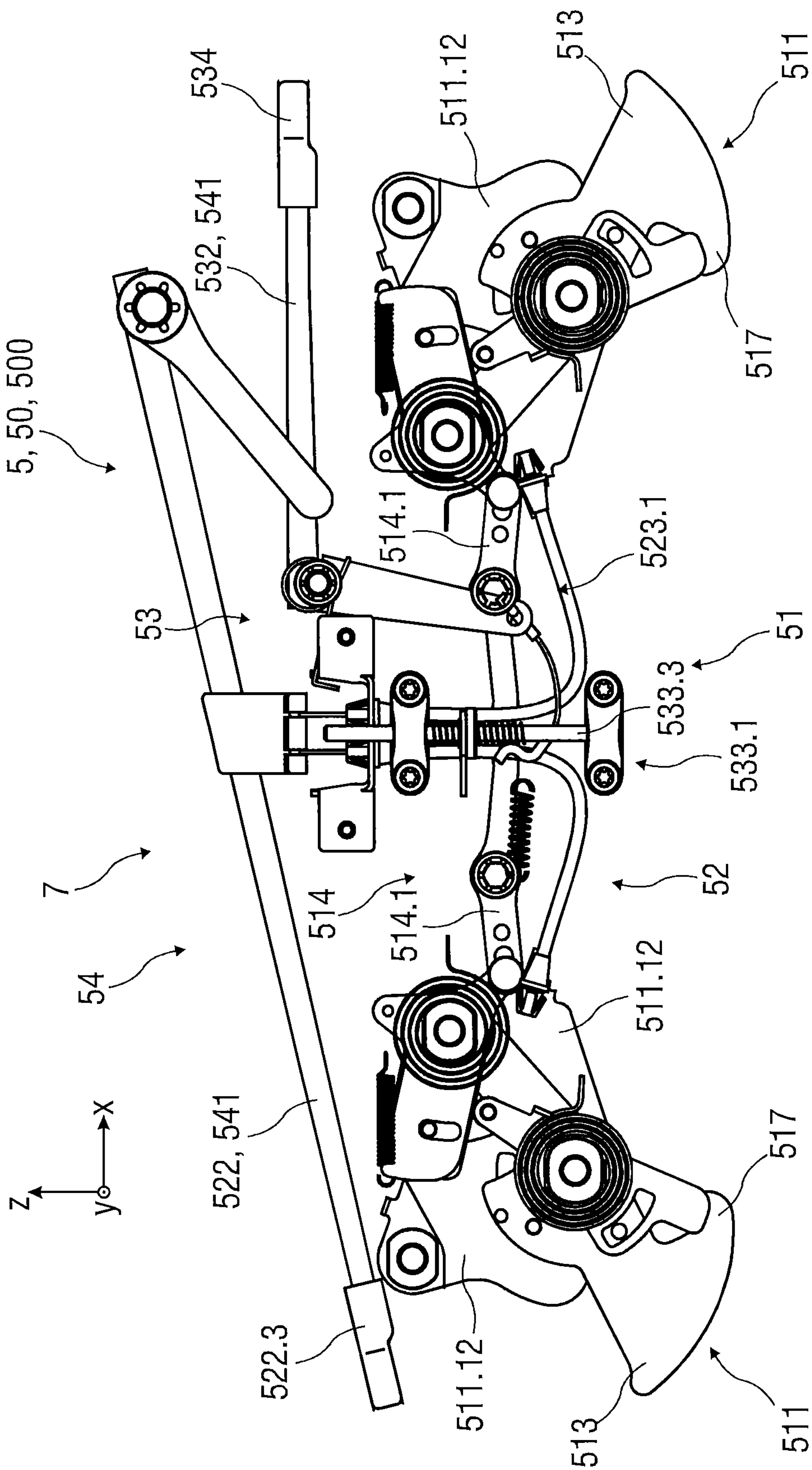

534
511.12
513
511
532, 541
5, 50, 500
517
514.1
523.1
53
51
533.3
533.1
7
514
52
54
514.1
511.12
z
x
y
522, 541
517
522.3
511.12
513
511

FIG 14A

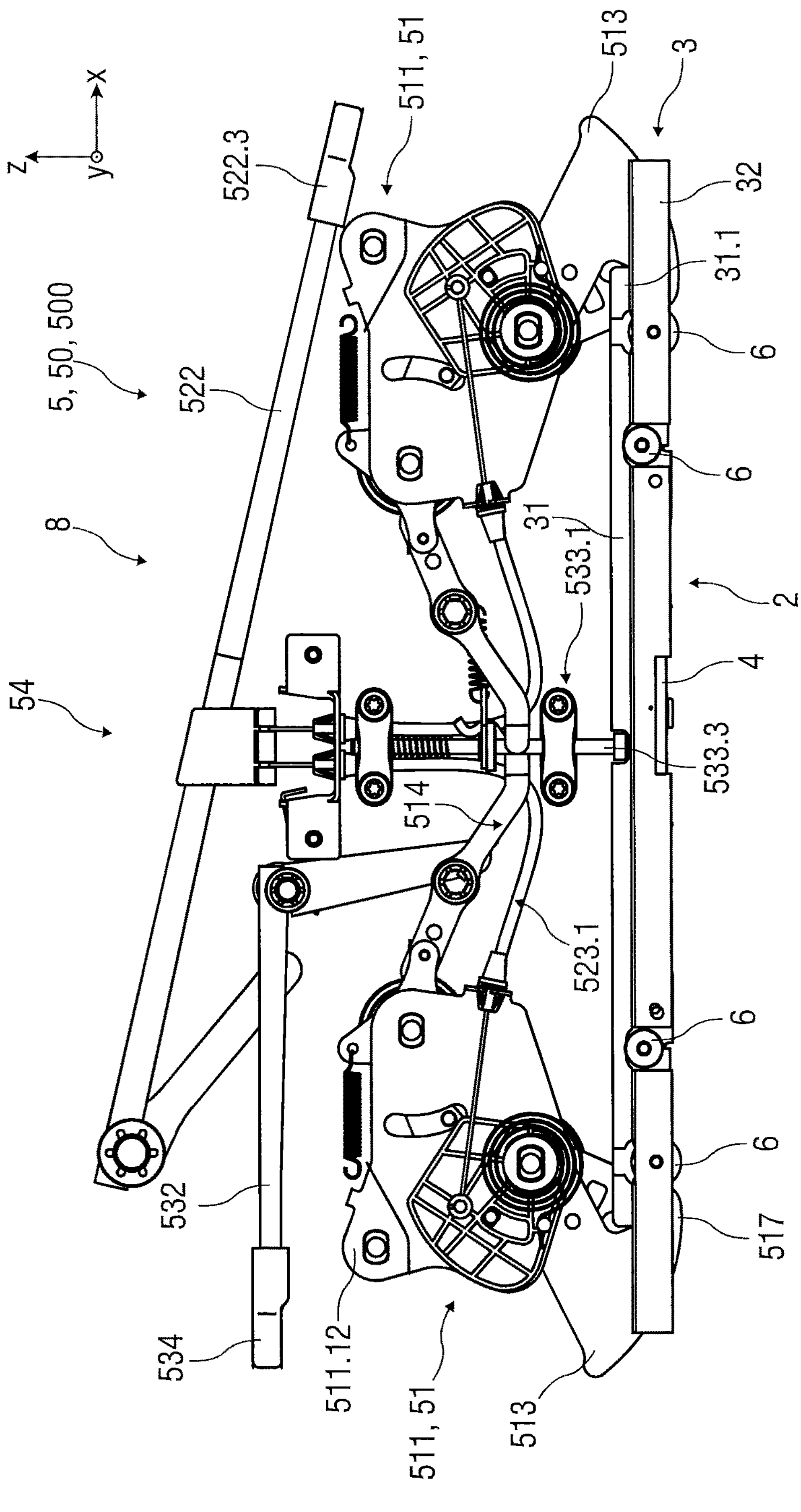

z
y
x
511, 51
513
3
522.3
32
31.1
5, 50, 500
522
6
6
8
31
533.1
2
54
4
533.3
514
523.1
6
6
532
517
534
511.12
511, 51
513

FIG 14B

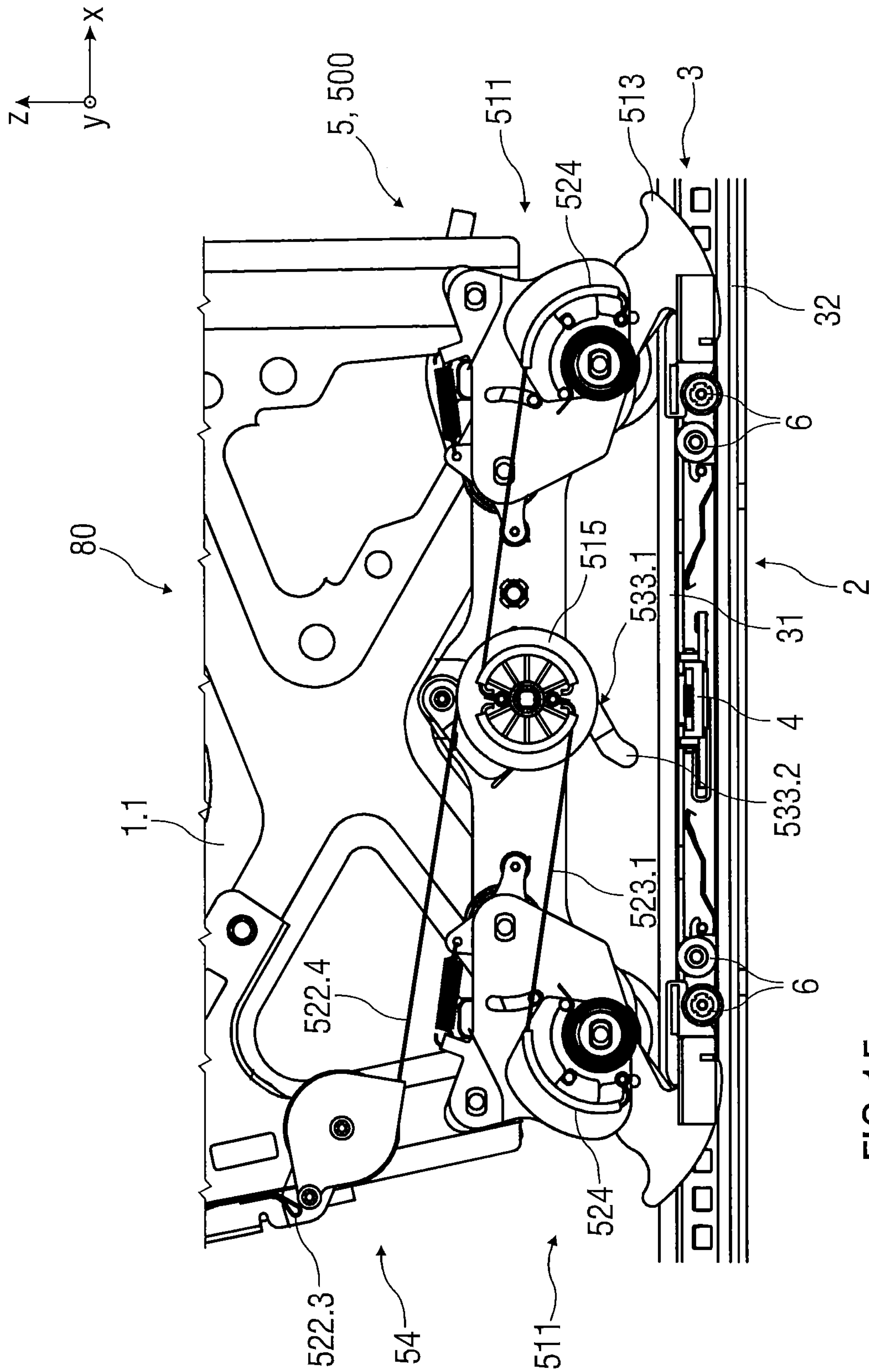

x
z
y
5, 500
511
524
513
3
32
6
80
515
533.1
2
31
4
533.2
1.1
523.1
522.4
6
524
522.3
54
511

FIG 15

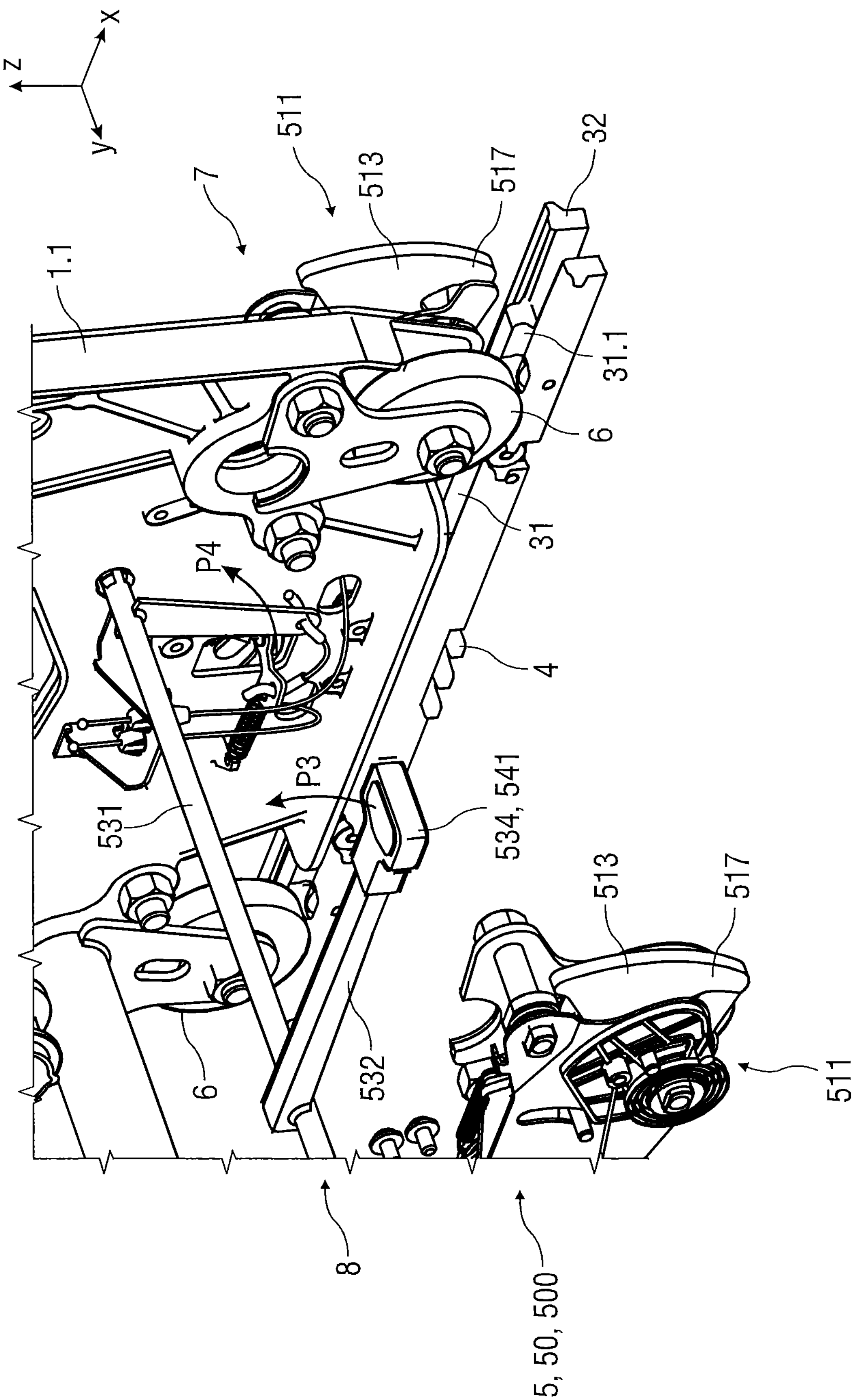

z
x
y
511
7
513
517
32
1.1
31.1
6
31
P4
4
P3
531
534, 541
513
517
6
532
511
8
5, 50, 500

FIG 16A

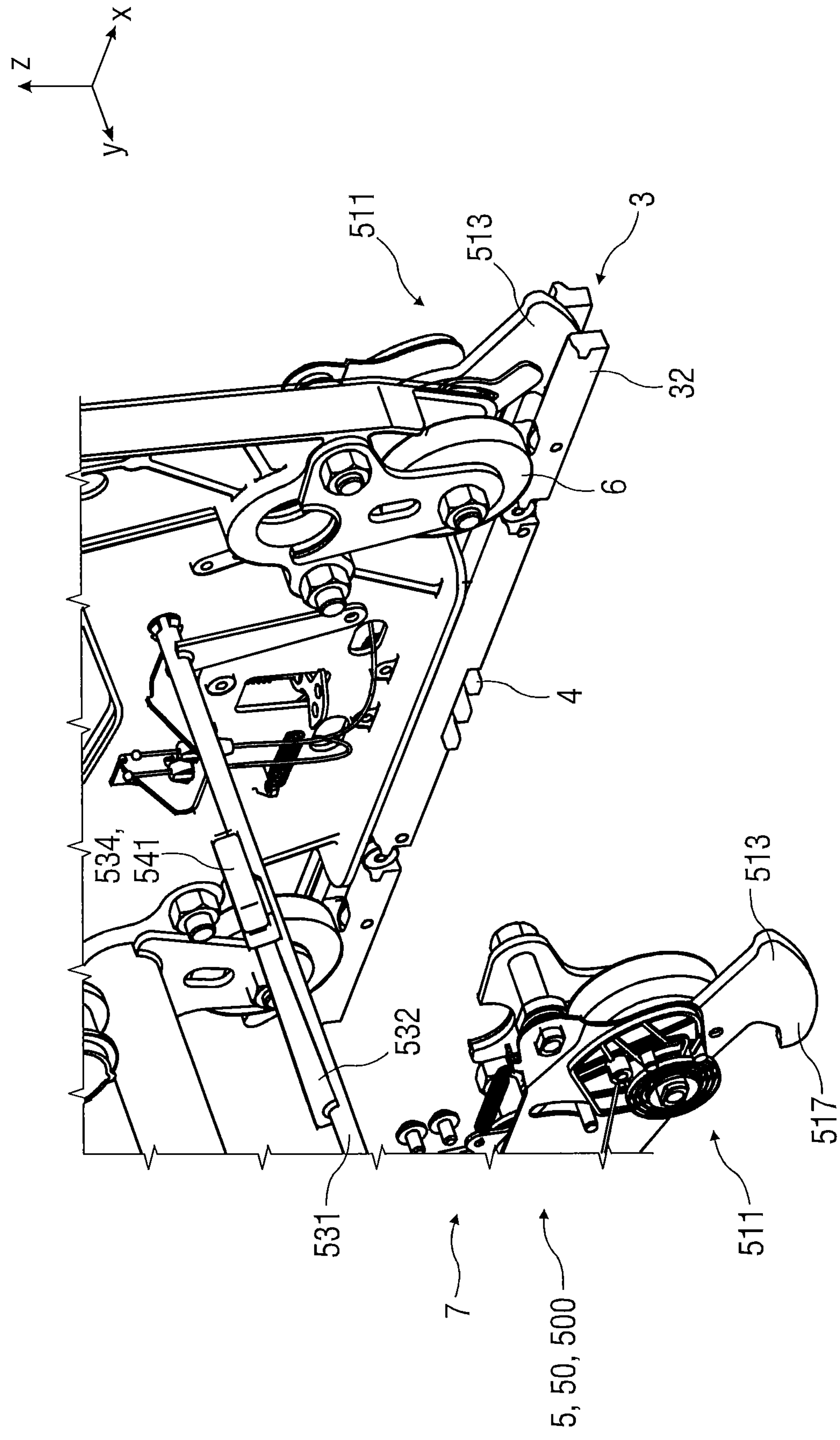
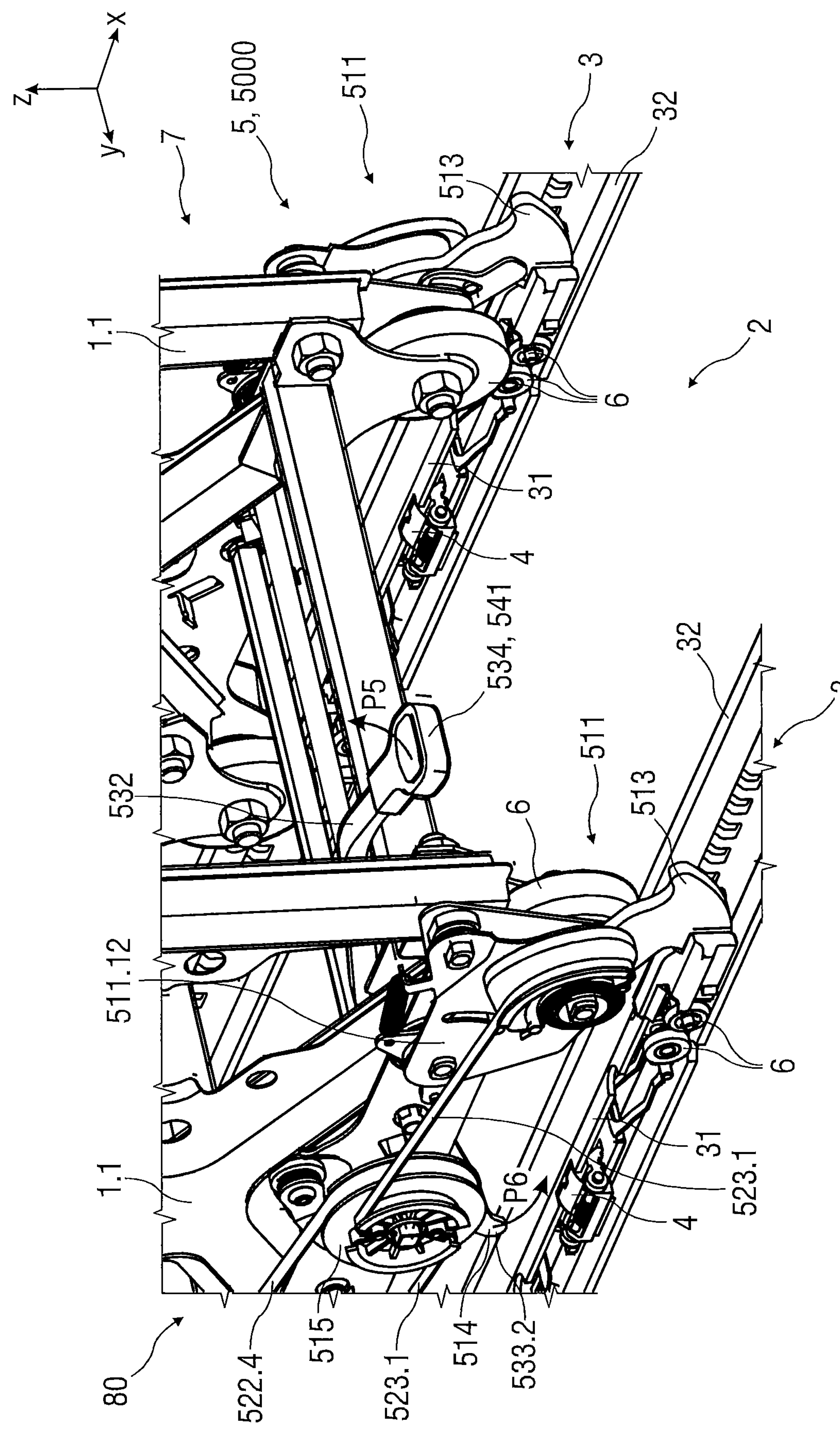


FIG 16B



x
z
y
7
5, 5000
511
3
32
513
1.1
2
6
31
4
534, 541
P5
532
511
3
32
513
6
511.12
6
31
523.1
1.1
P6
4
80
522.4
515
523.1
514
533.2

FIG 17A

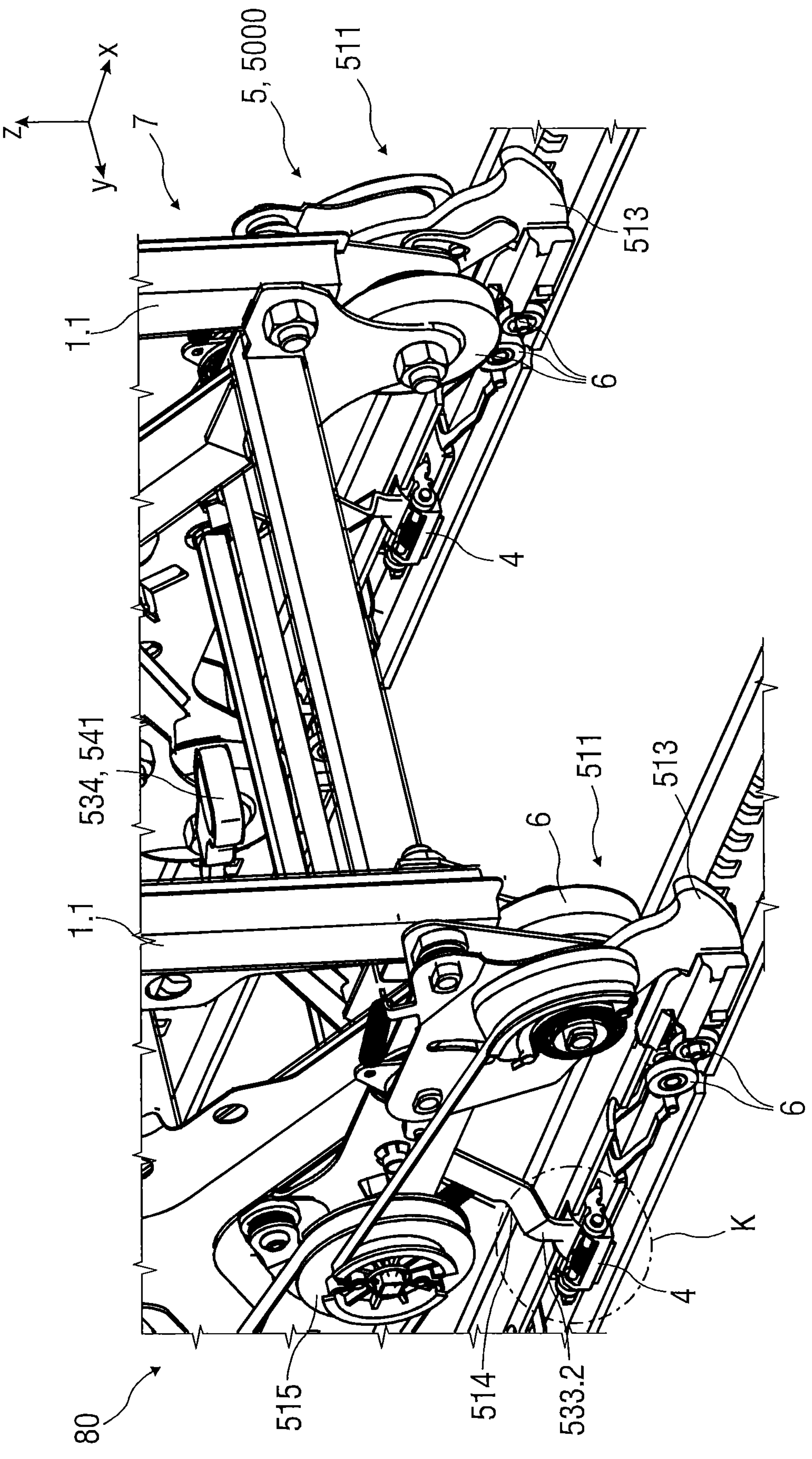


FIG 17B

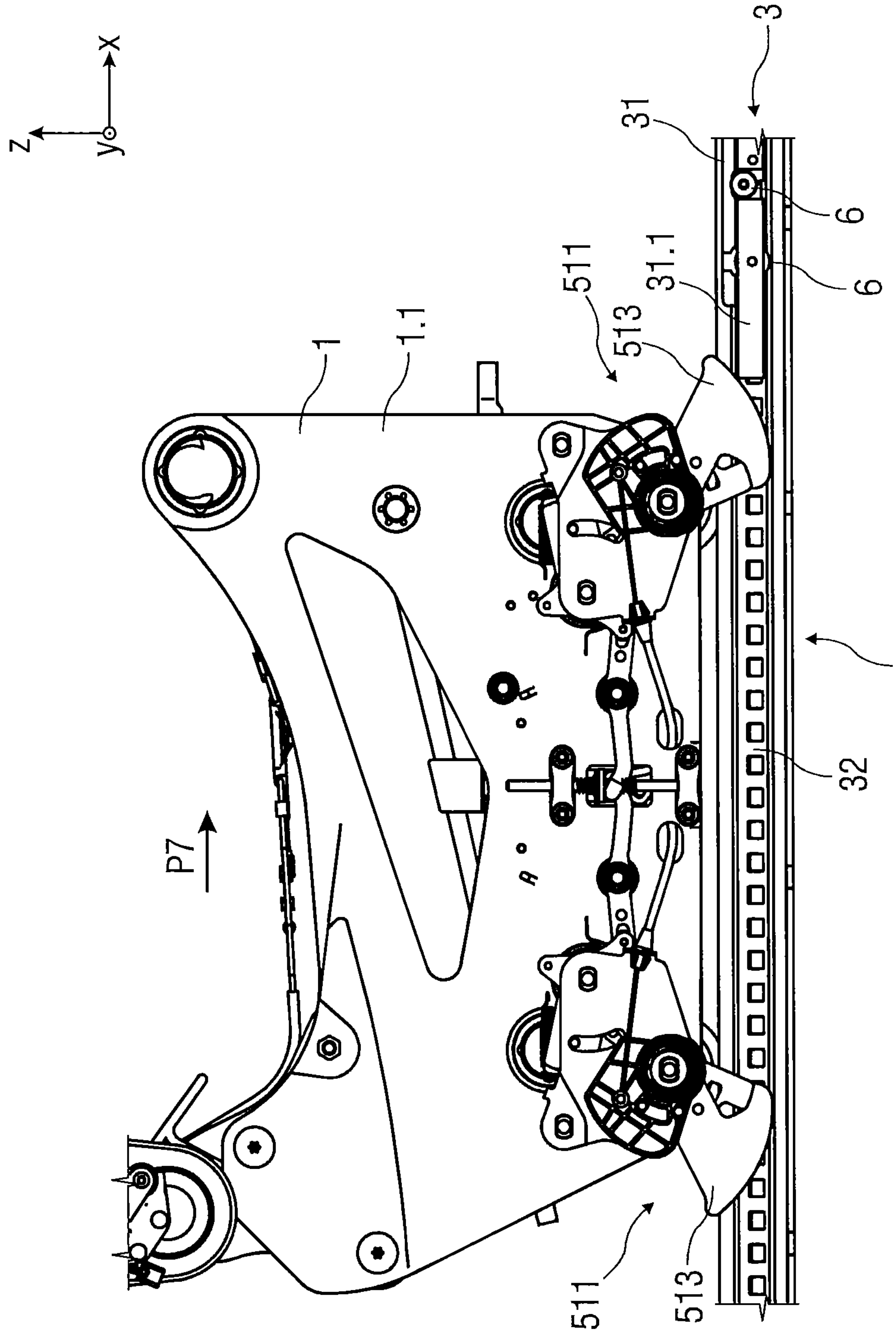


FIG 18A

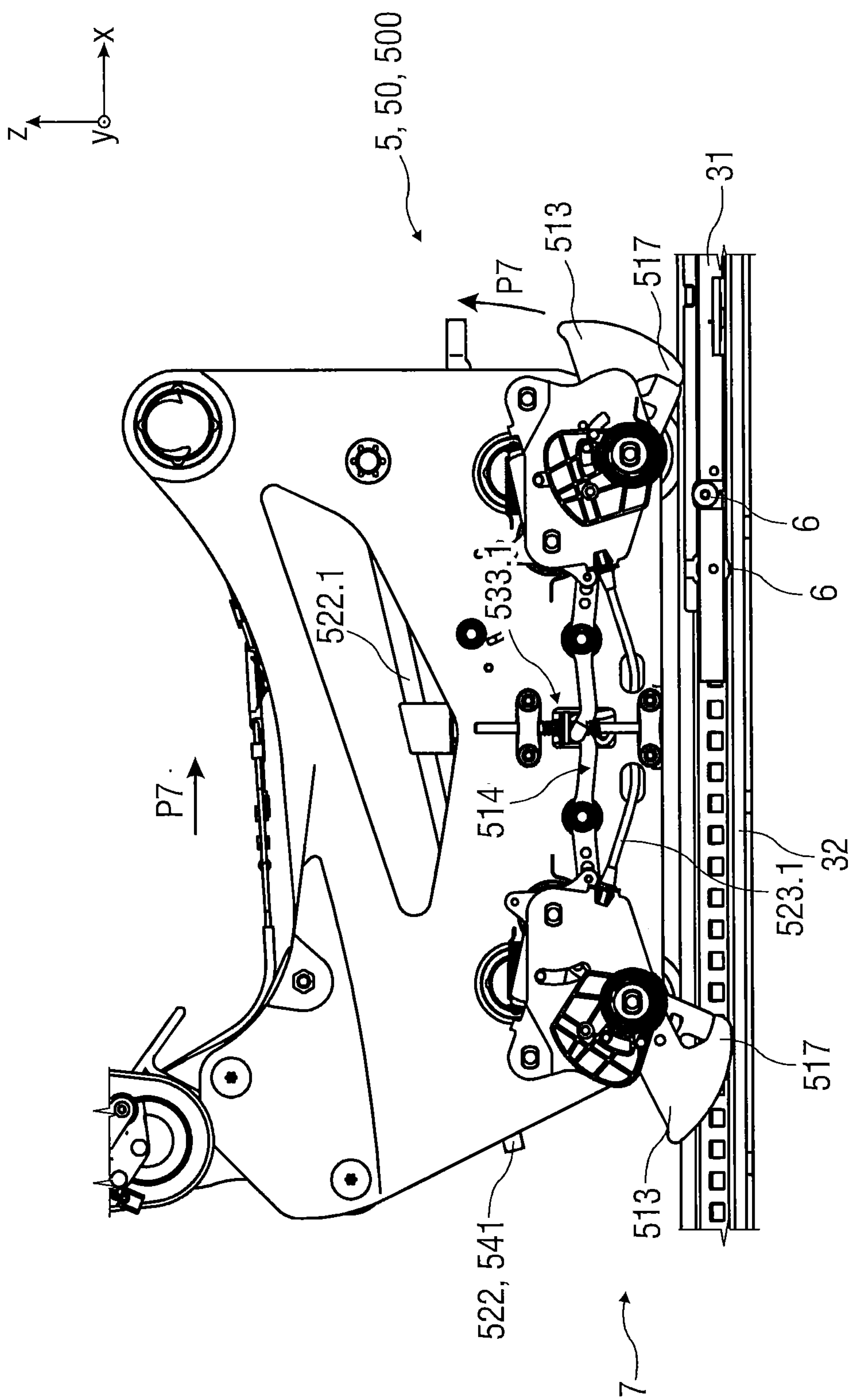
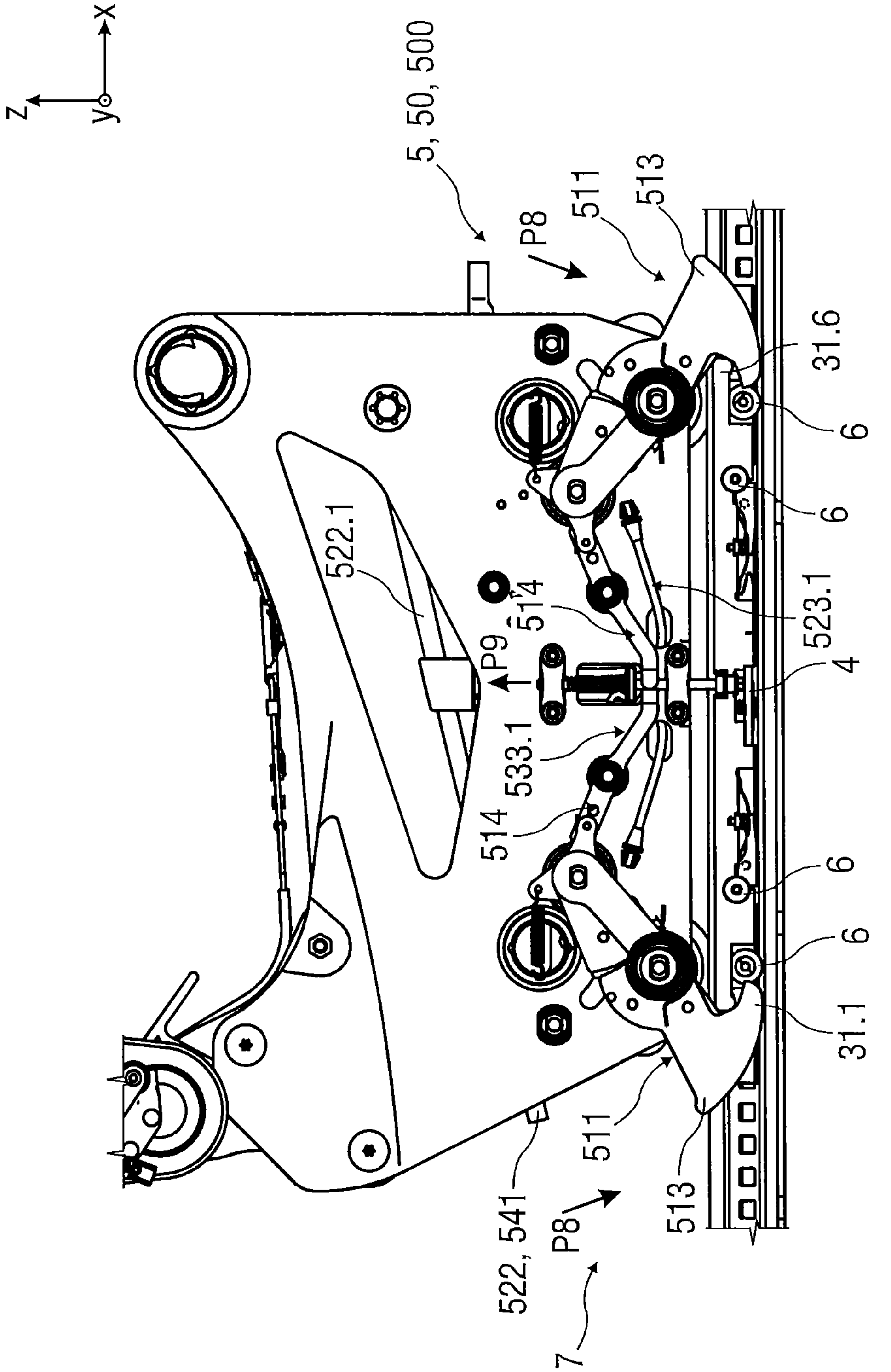
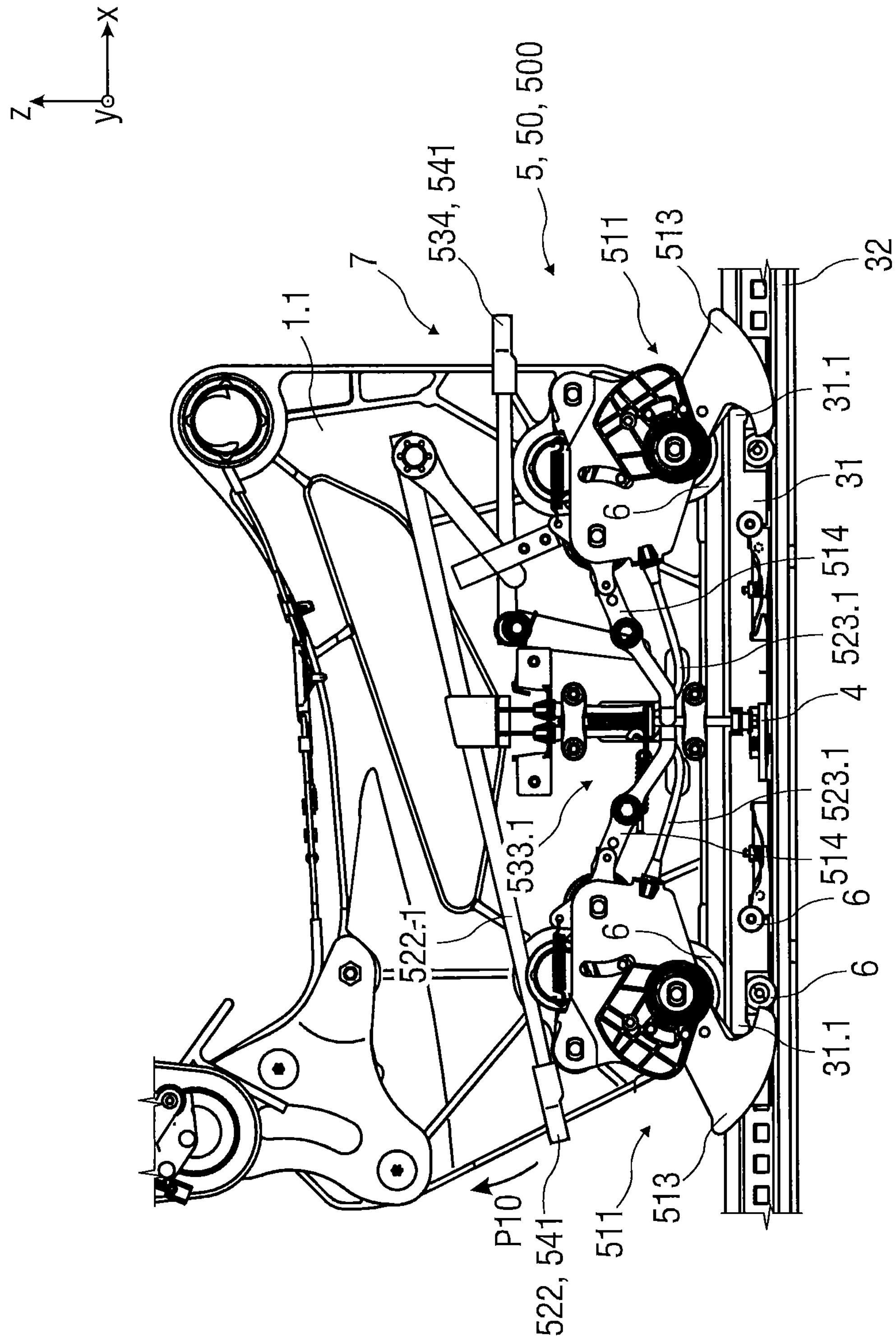

z
x
y
5, 50, 500
P7
513
517
31
6
6
522.1
533.1
P7
514
523.1
32
517
513
522, 541
7

FIG 18B



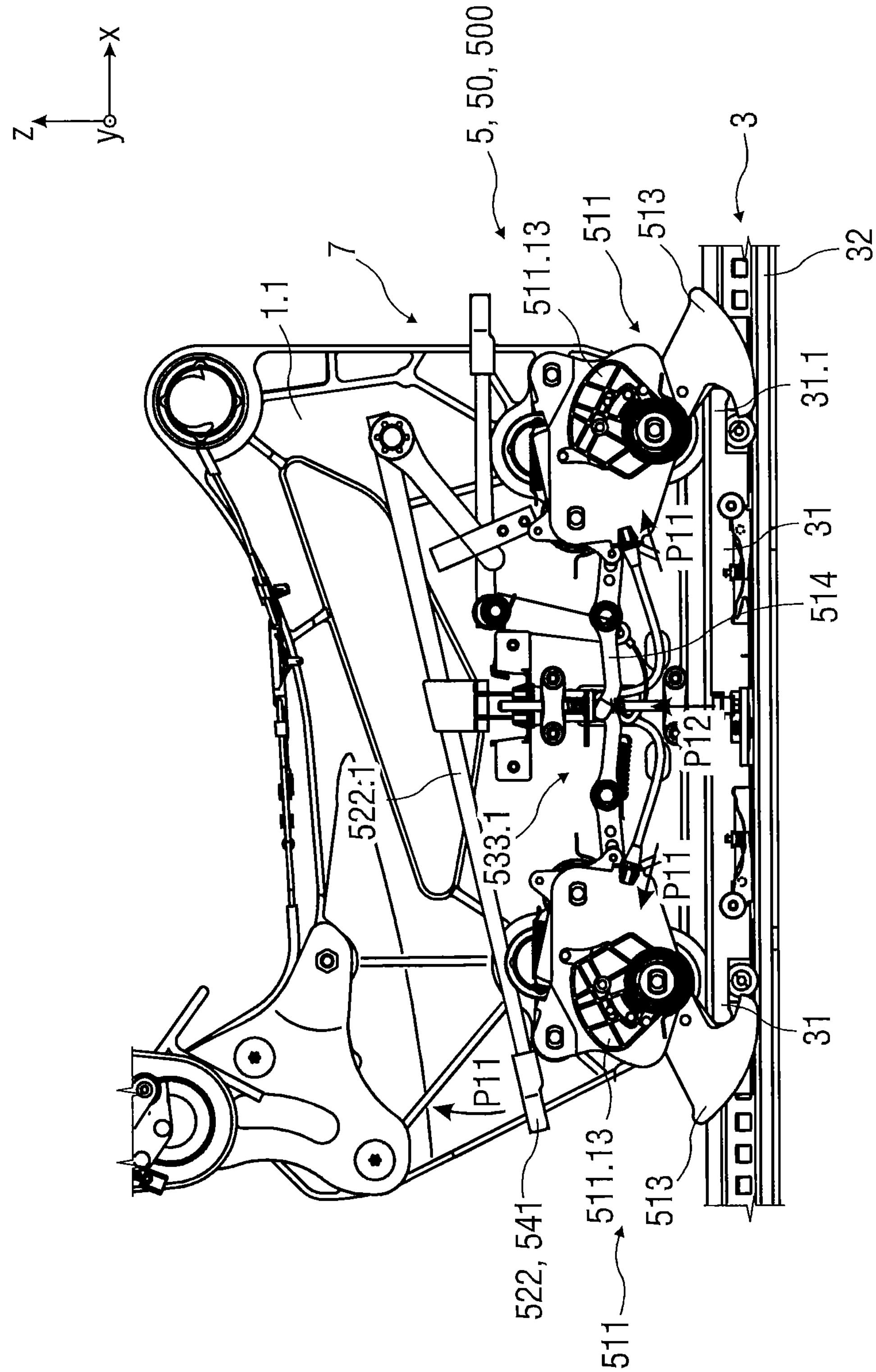
z
y
x
5, 50, 500
P8
511
513
31.6
6
6
522.1
514
523.1
P9
4
533.1
514
6
6
31.1
511
513
522, 541
P8
7

FIG 18C



x
z
y
5, 50, 500
534, 541
7
1.1
511
513
32
31.1
31
6
514
523.1
4
523.1
514
6
6
31.1
533.1
522.1
P10
522, 541
511
513

FIG 18D



z
x
y
5, 50, 500
7
1.1
511.13
511
513
3
32
31.1
31
514
P11
P12
522.1
533.1
522, 541

FIG 18E

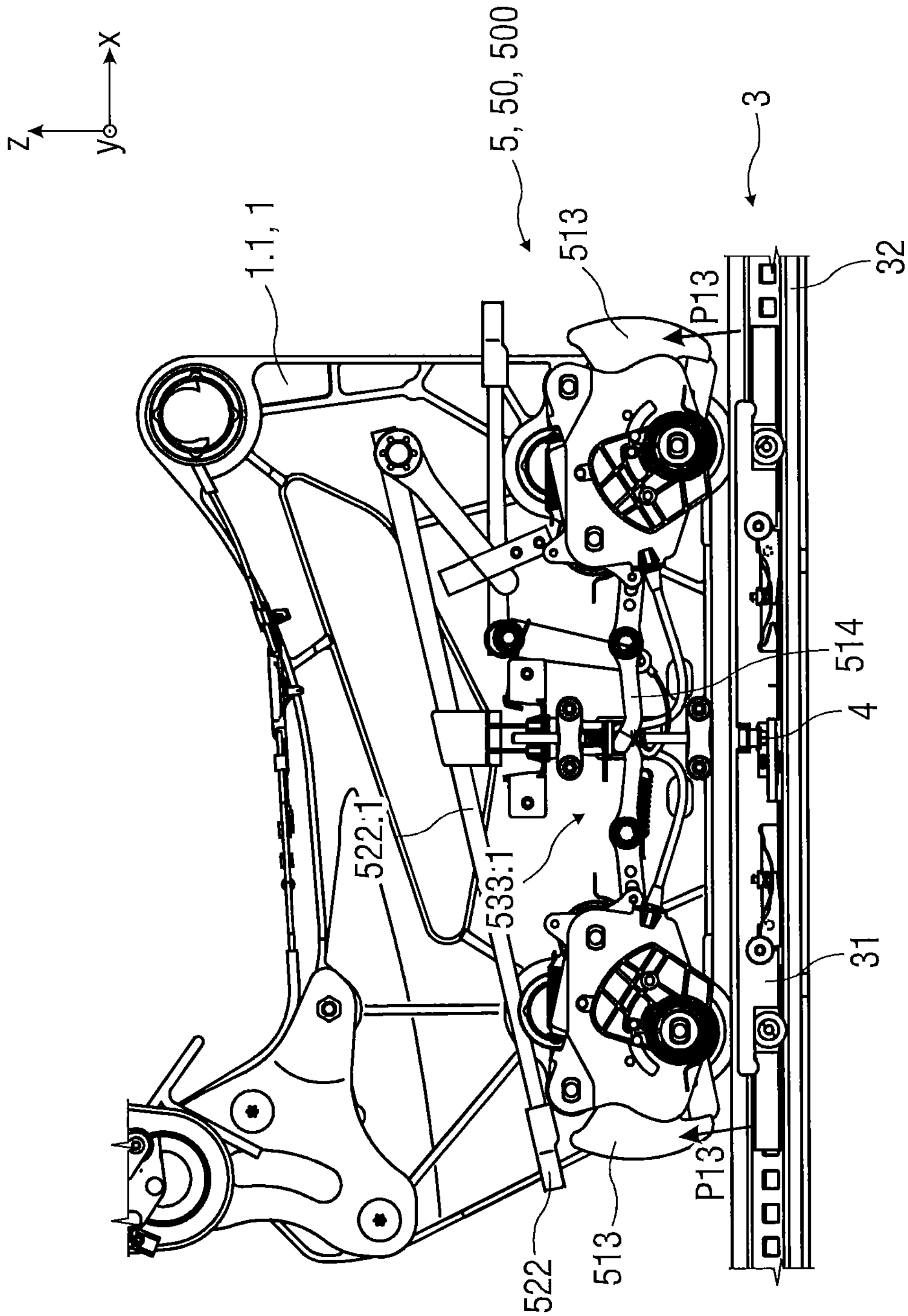
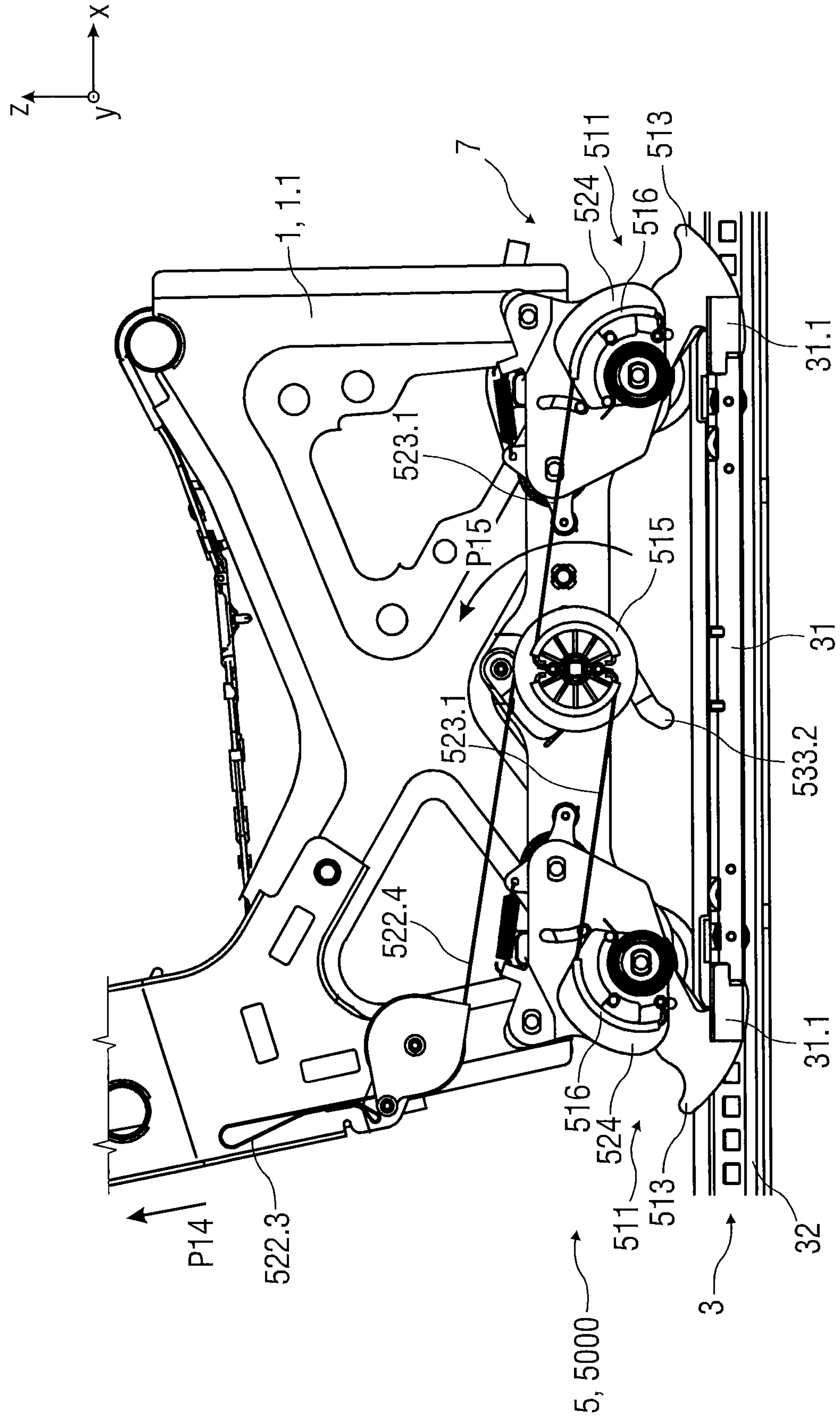


FIG 18F


x
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y
7
1, 1.1
524
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516
513
31.1
523.1
P15
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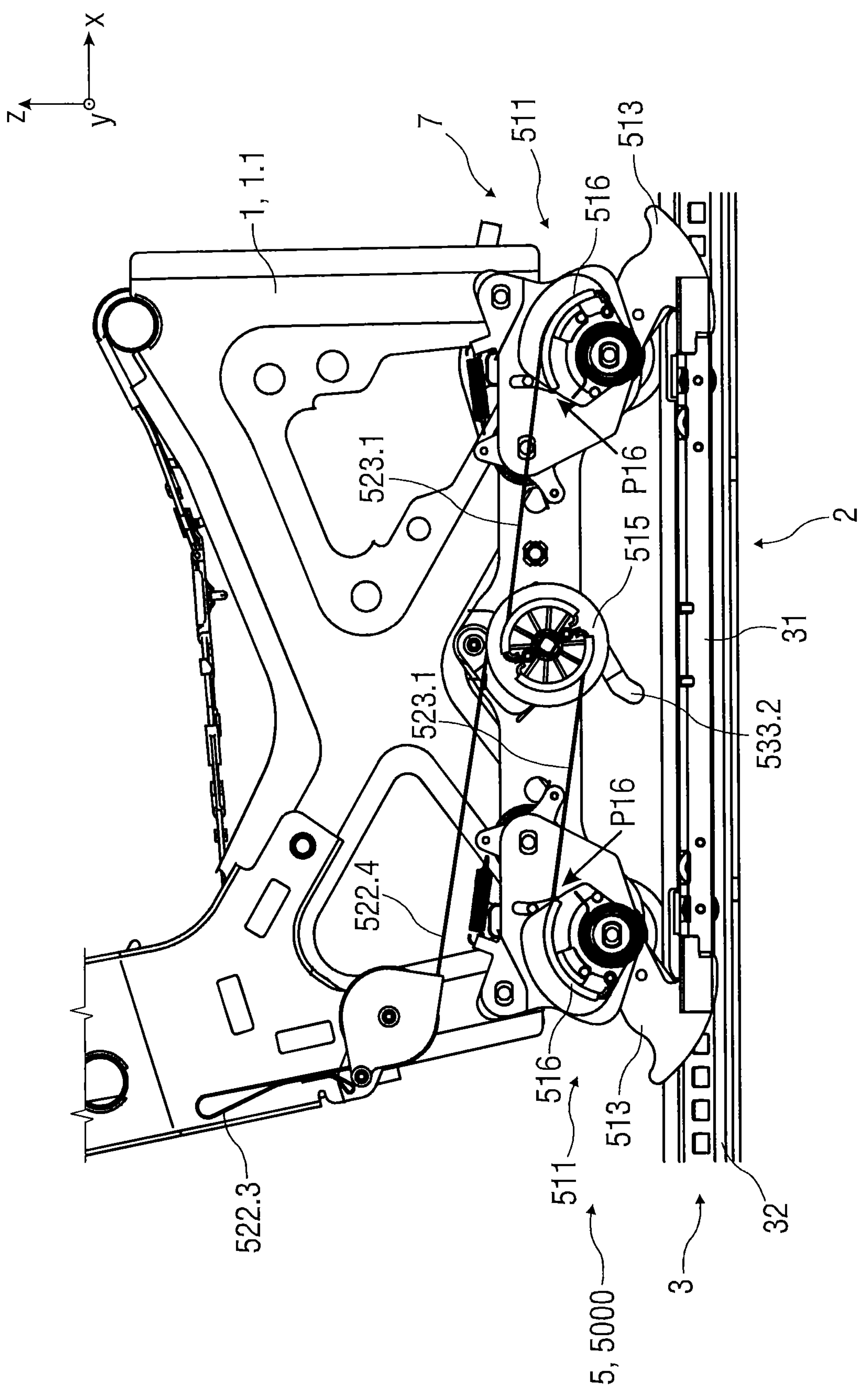
522.4
31.1
516
524
511
513
522.3
P14
32
3
5, 5000

FIG 19A


x
z
y
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523.1
P16
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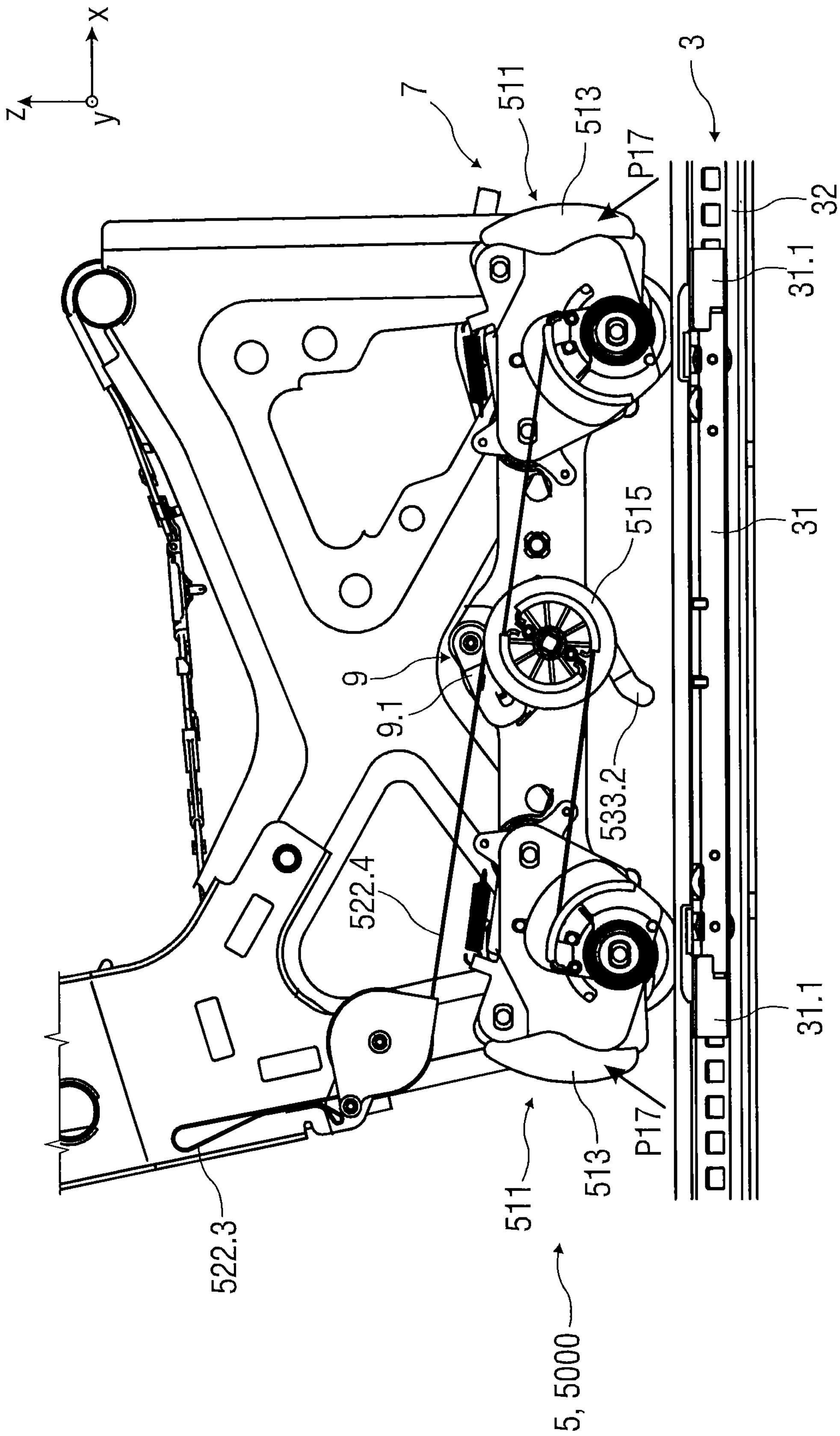
533.2
523.1
P16
522.4
516
513
511
522.3
5, 5000
3
32

FIG 19B


x
z
y
7
511
513
P17
3
32
31.1
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515
9
9.1
533.2
522.4
31.1
P17
513
511
522.3
5, 5000

FIG 19C

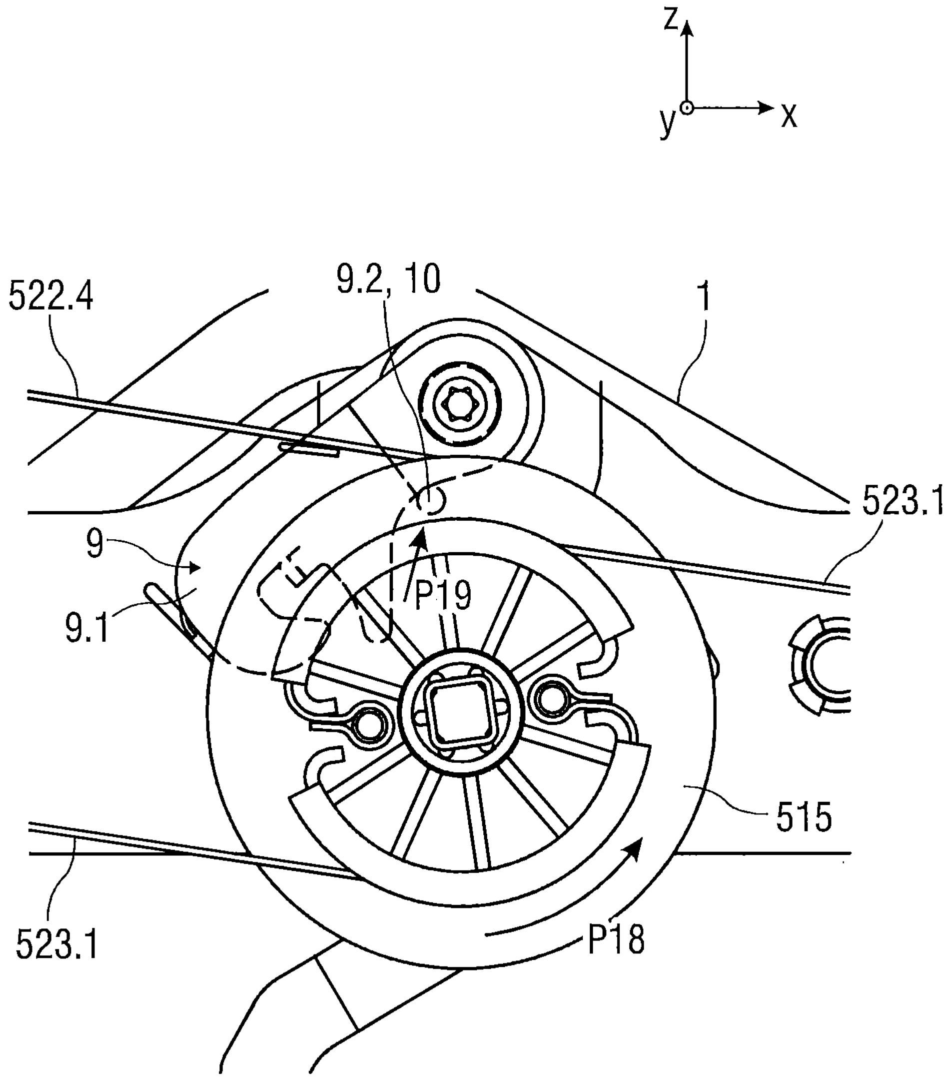

FIG 20A

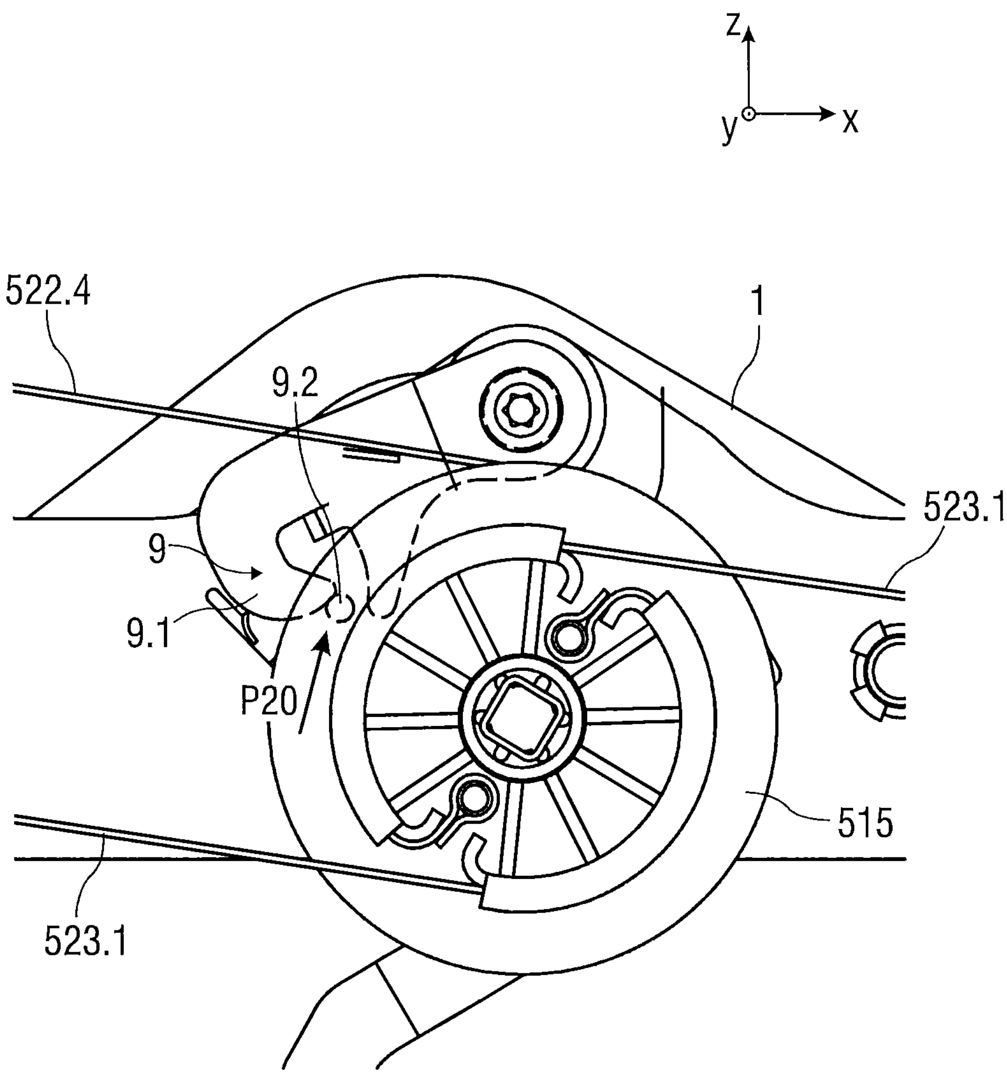

FIG 20B

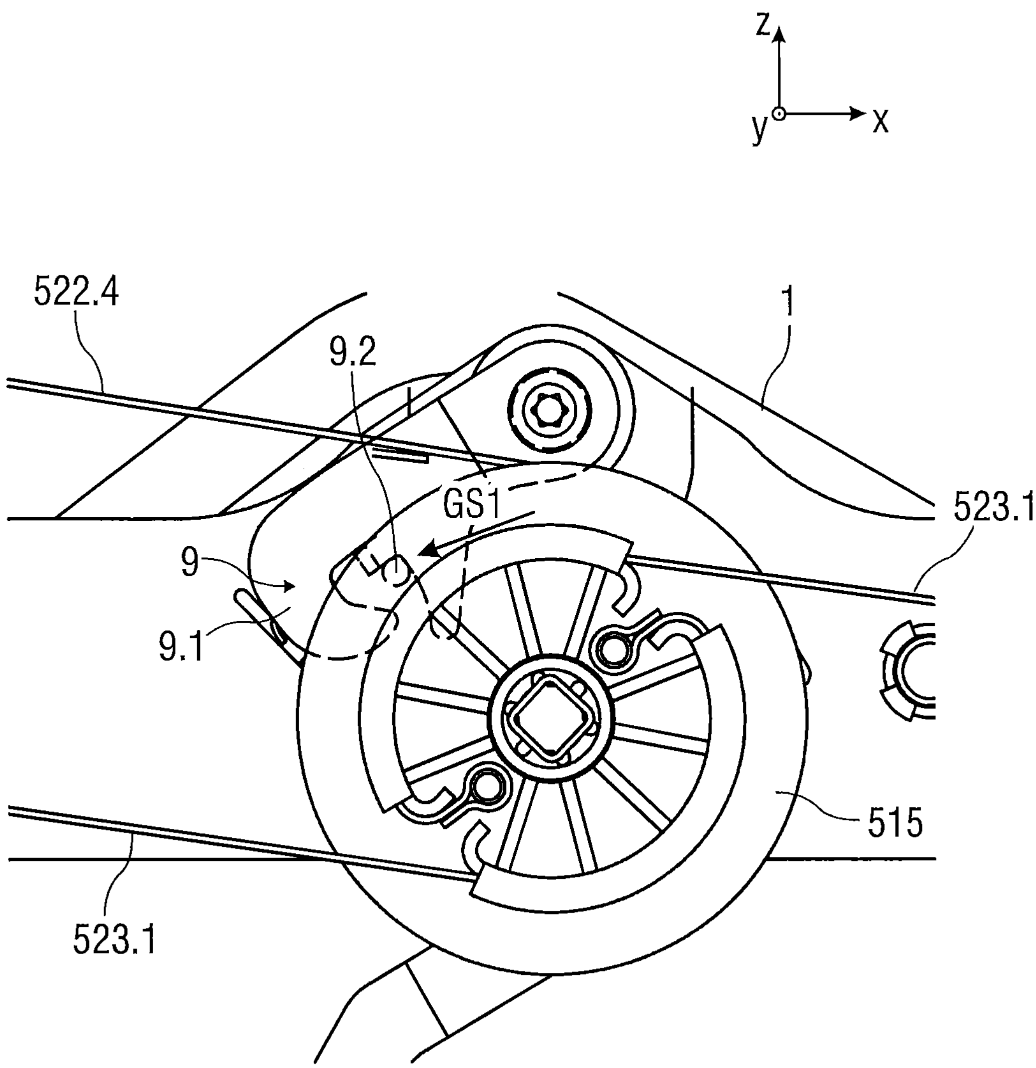

FIG 20C

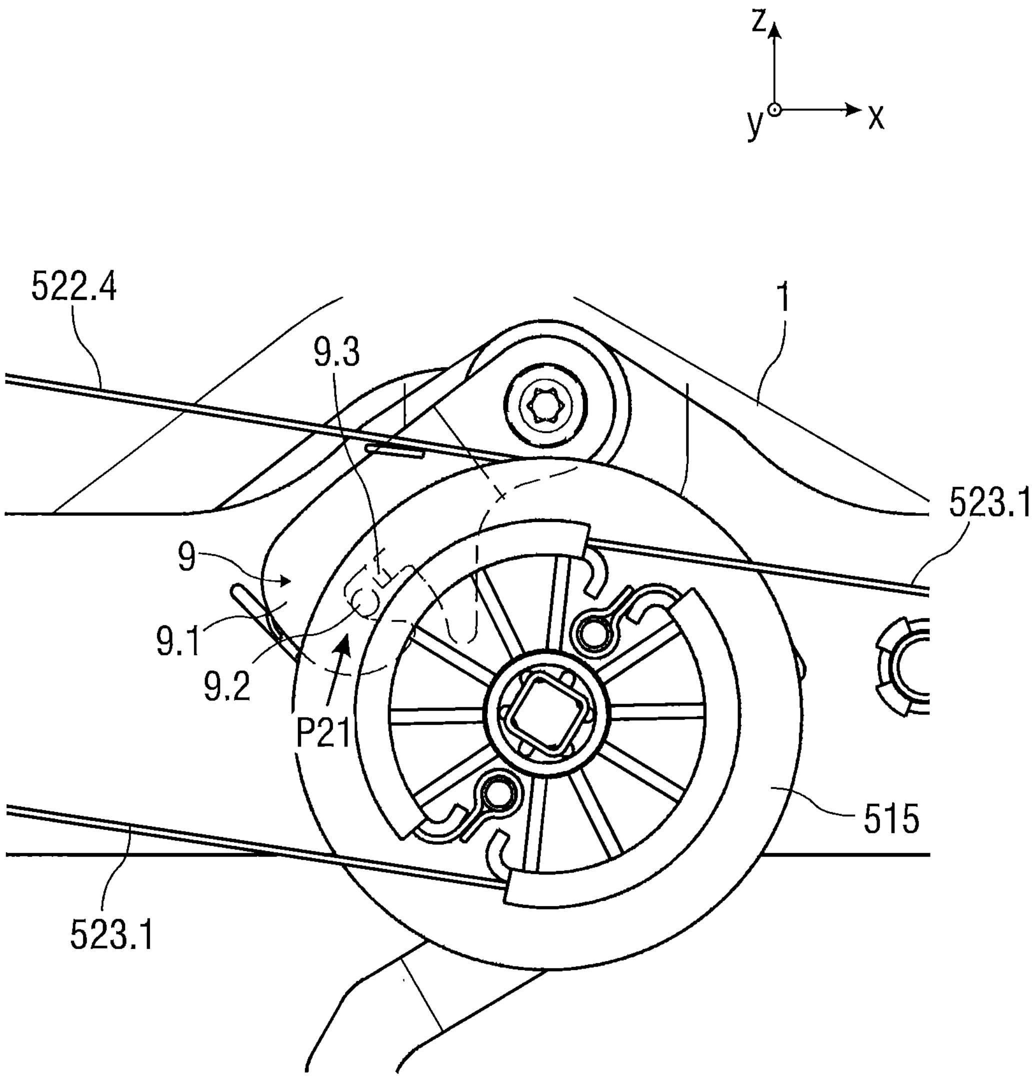

FIG 20D

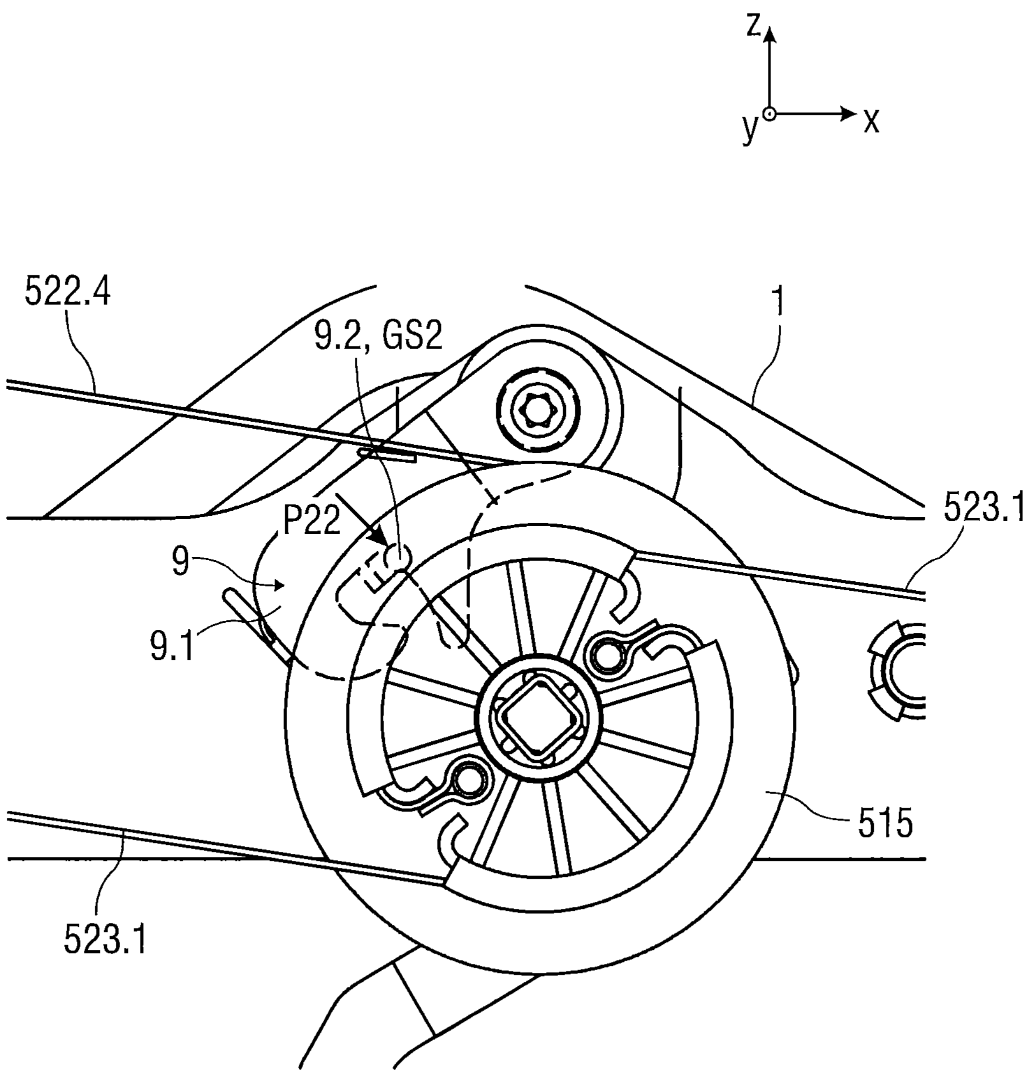

FIG 20E

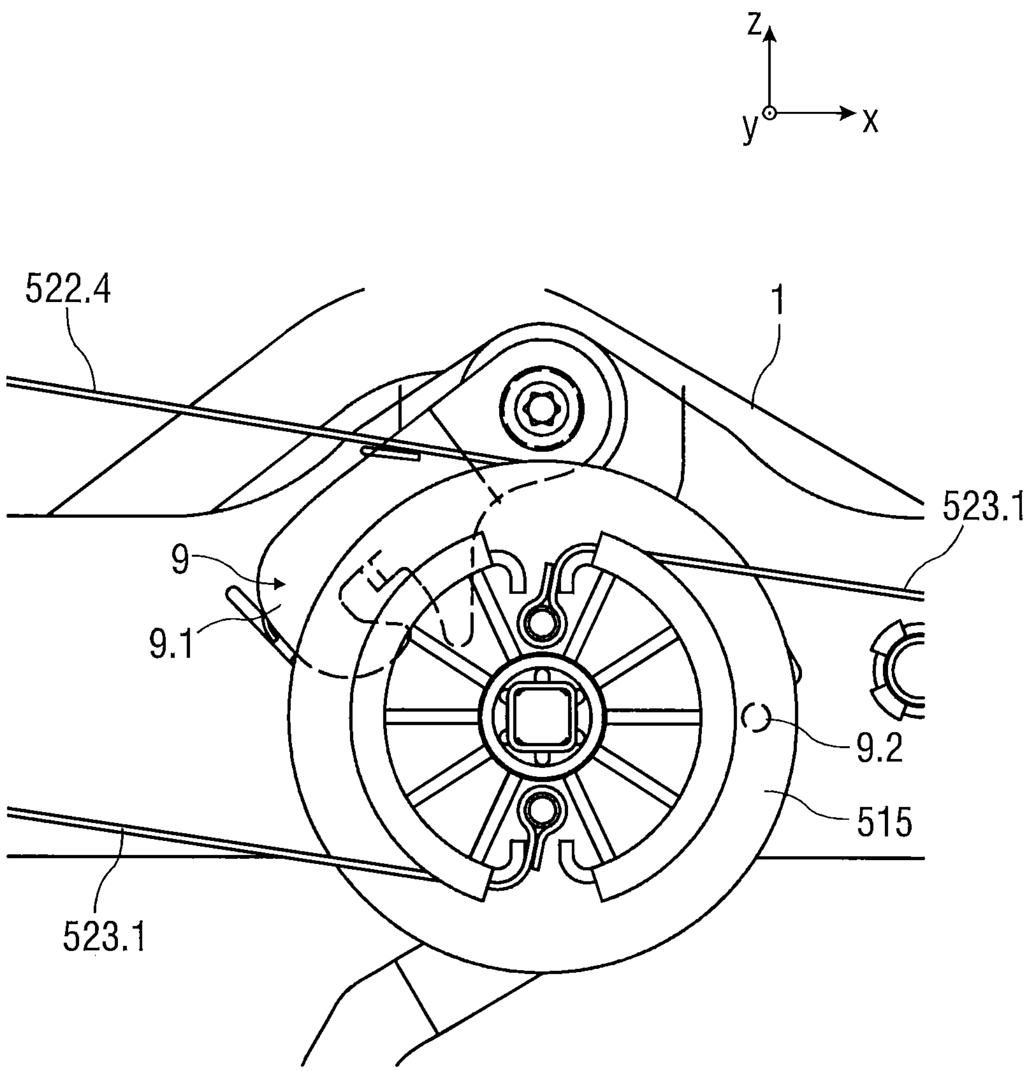

FIG 20F

FASTENING ARRANGEMENT AND SEAT

FIELD

The invention relates to a fastening arrangement of a seat on a rail arrangement and to a seat with such a fastening arrangement.

BACKGROUND

Fastening arrangements for a seat are known in the prior art. For example, the seat can be fastened on a rail arrangement releasably by hooks or screws or nonreleasably by a welded connection.

SUMMARY

It is the object of the present invention to specify a simple fastening arrangement of a seat on a rail arrangement and a seat with such a simple fastening arrangement.

The object is achieved according to the invention by a fastening arrangement of a seat on a rail arrangement, wherein the fastening arrangement is of modular construction and comprises at least one lock module for releasably arresting the fastening arrangement on the rail arrangement, one lock unlocking module and one rail unlocking module.

Using such a modular seat fastening arrangement, a seat can be simply and rapidly mounted on and removed from a seat longitudinal adjustment device. The individual modules, such as the lock module, the lock unlocking module and the rail unlocking module, can be preassemblable here. Such a modular fastening arrangement has few parts and can be mounted particularly simply and rapidly and also has a low weight.

Furthermore, the lock unlocking module and the rail unlocking module are operable in a simple manner independently of one another or synchronously, i.e. can be brought from an arresting position of a seat lock and/or a rail locking means into a released position of the seat lock or of the rail locking means, or vice versa.

The lock module can be of multi-part design and comprises, for example, two or four lock units, e.g. one pair of lock units or two pairs of lock units. In one possible embodiment, the lock units are synchronously operable, in particular unlockable, by the lock unlocking module.

The lock units are preferably arranged on a common carrier. In particular, the lock units can be preassembled on the common carrier.

Furthermore, the fastening arrangement can comprise an actuating module for actuating the lock unlocking module and/or the rail unlocking module. The actuating module can comprise one or more actuating units for dependently or independently actuating the lock unlocking module and/or the rail unlocking module. The actuating units can be formed separately. However, the actuating units can also be coupled in terms of movement. The actuating units can be designed as a lever arrangement, in particular as a lever arm arrangement, and/or as a tension arrangement, in particular tension strap arrangement or tension belt arrangement.

In one possible embodiment, the actuating module is coupled to the lock module via the lock unlocking module.

Alternatively or additionally, the actuating module can be coupled to a locking element for the rail arrangement via the lock unlocking module.

With regard to the seat according to the invention, the object is achieved by a rail arrangement and the previously described fastening arrangement, which is designed for releasably fastening the seat on the rail arrangement.

In one possible embodiment, the fastening arrangement is designed as a preassemblable docking mechanism. The docking mechanism here comprises the lock module and also the lock unlocking module and the rail unlocking module. Preferably, an unlocking element and/or an unlocking mechanism for the rail arrangement can additionally be part of the preassemblable docking mechanism. Such a modular preassemblable docking mechanism for releasably fastening the seat on the rail arrangement permits simple and rapid installation or removal of the seat on or from the rail arrangement.

In one possible embodiment, the rail arrangement comprises two pairs of rails each having an upper rail and a lower rail. The fastening arrangement is releasably arrestable at one rail end of the upper rail by a respective lock unit. Such an in particular end-face releasable arresting of the fastening arrangement on the rail ends of the upper rail is simply constructed and permits simple and reliable installation of the fastening arrangement on the rail arrangement.

DESCRIPTION OF THE FIGURES

Exemplary embodiments of the invention will be discussed in more detail with reference to drawings, in which:

FIG. **1** shows a perspective illustration of a seat with a rail arrangement,

FIG. **2** shows a schematic illustration of an embodiment of a modular fastening arrangement of a seat on a rail arrangement,

FIG. **3** schematically shows, in an exploded illustration, a first embodiment of a modular fastening arrangement,

FIG. **4** schematically shows, in an exploded illustration, a second embodiment of a modular fastening arrangement,

FIG. **5** schematically shows, in a perspective view, a second embodiment of a modular fastening arrangement,

FIG. **6** schematically shows, in an exploded illustration, a third embodiment of a modular fastening arrangement,

FIG. **7** schematically shows, in a perspective view, a third embodiment of a modular fastening arrangement,

FIG. **8** schematically shows, in an exploded illustration, an exemplary embodiment of a lock unit,

FIG. **9** schematically shows, in the assembled state, an exemplary embodiment of a lock unit,

FIG. **10** shows a schematic illustration of a releasable seat on a rail arrangement,

FIG. **11** shows a schematic illustration of the releasable seat pre-positioned on the rail arrangement, and

FIG. **12** shows a schematic perspective view of one embodiment of a fastening arrangement in the unlocked or open state,

FIG. **13** shows a schematic perspective view of one embodiment of an actuating module,

FIG. **14**A shows a schematic side view of one embodiment of a carrier plate of a lock unit,

FIG. **14**B shows a schematic side view of one embodiment of an unlocking mechanism on the carrier plate of a lock unit,

FIG. **15** shows a schematic side view of one embodiment of an alternative unlocking mechanism,

FIG. **16**A shows a schematic perspective view of one embodiment of blocking elements,

FIG. **16**B shows a schematic perspective view of one embodiment of blocking elements,

FIG. **17**A shows a schematic perspective view of one embodiment of blocking elements, FIG. **17**B shows a schematic perspective view of one embodiment of blocking elements,

FIG. **18**A shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **18**B shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **18**C shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **18**D shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **18**E shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **18**F shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,

FIG. **19**A shows a schematic side view of one embodiment of a cable mechanism in a movement sequence,

FIG. **19**B shows a schematic side view of one embodiment of the cable mechanism in a movement sequence,

FIG. **19**C shows a schematic side view of one embodiment of the cable mechanism in a movement sequence,

FIG. **20**A shows a schematic side view of one embodiment of a cable arresting means in a movement sequence,

FIG. **20**B shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

FIG. **20**C shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

FIG. **20**D shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

FIG. **20**E shows a schematic side view of one embodiment of the cable arresting means in a movement sequence, and

FIG. **20**F shows a schematic side view of one embodiment of the cable arresting means in a movement sequence.

DETAILED DESCRIPTION

Parts which correspond to one another are denoted by the same reference designations in all figures.

FIG. **1** shows schematically, in a perspective illustration, a seat **1**, in particular vehicle seat, with a seat longitudinal adjustment device. The seat **1** is arranged, for example, on a vehicle body, not illustrated specifically, by the seat longitudinal adjustment device.

A vehicle seat **1** which is schematically illustrated in FIG. **1** is described below using three spatial directions which run perpendicularly to one another. In the case of a seat **1** installed in the vehicle, a longitudinal direction X runs largely horizontally and preferably parallel to a vehicle longitudinal direction which corresponds to the normal direction of travel of the vehicle. A transverse direction Y which runs perpendicularly to the longitudinal direction X is likewise oriented horizontally in the vehicle and runs parallel to a vehicle transverse direction. A vertical direction Z runs perpendicularly to the longitudinal direction X and perpendicularly to the transverse direction Y. In the case of a vehicle seat **1** which is installed in the vehicle, the vertical direction Z runs preferably parallel to a vehicle vertical axis.

The positional and directional indications used, such as for example front, rear, top and bottom, relate to a viewing direction of an occupant seated in the seat **1** in a normal seating position, wherein the seat **1** is installed in the vehicle, in a use position suitable for passenger transport, with an upright backrest, and is oriented in the conventional manner in the direction of travel. The seat **1** may also be installed or moved, however, in a differing orientation, for example transversely with respect to the direction of travel.

The seat longitudinal adjustment device is designed as a rail arrangement **2**. The rail arrangement **2** comprises, for example, two pairs of rails **3** which are arranged at a distance from one another. The respective pair of rails **3** comprises an upper rail **31**, also called rail runner or seat rail, and a lower rail **32**, also called guide rail or floor rail. The upper rail **31** is arranged on the lower rail **32** so as to be longitudinally adjustable between a front end position and a rear end position. This adjustability permits a longitudinal adjustment of the position of the seat **1** in the longitudinal direction X, with the frontmost position of the seat **1** being assigned to the front end position of the upper rail **31** and the rearmost position of the seat **1** being assigned to the rear end position of the upper rail **31**. The upper rail **31** and the lower rail **32** each have associated rail ends **33**.

The rail arrangement **2** furthermore comprises at least one locking element **4**. Each pair of rails **3** here can comprise an associated locking element **4**, for example in the form of a latching plate or locking plate.

For the releasable fastening of the seat **1** on the rail arrangement **2**, various embodiments of a modular fastening arrangement **5** are described below. The respective fastening arrangement **5** can be preassembled. In particular, the fastening arrangement **5** can be designed as a preassemblable docking mechanism **7**. Such a modular preassemblable docking mechanism **7** permits simple and rapid installation or removal of the seat **1** on or from the rail arrangement **2**.

FIG. **2** shows a schematic illustration of the modular fastening arrangement **5** of the seat **1** for fastening the seat **1** on the rail arrangement **2**. The fastening arrangement **5** comprises at least one lock module **51** for releasably arresting the fastening arrangement **5** on the rail arrangement **2**, one lock unlocking module **52** for unlocking the lock module **51** and one rail unlocking module **53** for unlocking the locking element **4**.

Such a modular fastening arrangement **5** designed in particular as a docking mechanism **7** can be used to simply and rapidly mount and remove the seat **1** on and from the seat longitudinal adjustment device, in particular on and from the rail arrangement **2**. The individual modules, such as the lock module **51**, the lock unlocking module **52** and the rail unlocking module **53**, can be preassemblable here. The lock module **51** in turn can comprise a plurality of separate lock units **511**. The separate lock units **511** can be coupled in terms of movement; in particular, they can be actuable independently of one another and/or dependently on one another.

Such a modular fastening arrangement **5** has few preassembled installation parts and is mountable particularly simply and rapidly. Furthermore, the lock unlocking module **52** and the rail unlocking module **53** are operable in a simple manner independently of one another or synchronously, in particular can be brought from an arresting position of a seat lock, in particular the lock units **511** of the lock module **51**, and/or of a rail locking means, in particular the locking element **4**, into a released position of the seat lock or of the rail locking means, or vice versa.

The lock module **51** can be of multi-part design and comprises, for example, four lock units **511**, e.g. two pairs of lock units **511**. In one possible embodiment, the lock units **511** are operable or actuable synchronously or independently of one another using the lock unlocking module **52**.

Furthermore, the various fastening arrangements **5** can comprise an actuating module **54** for actuating the associated lock unlocking module **52** and the rail unlocking module **53**. The actuating module **54** can comprise one or more actuating units **541** for independently or dependently actuating the lock unlocking module **52** and/or the rail unlocking module **53**. The actuating units **541** can be formed separately or can be an integral part of the lock unlocking module **52** and/or of the rail unlocking module **53**. However, the actuating units **541** can also be coupled in terms of movement. The actuating units **541** can be designed as a lever arrangement, in particular as a lever arm arrangement, and/or as a tension arrangement, in particular tension strap arrangement or tension belt arrangement.

FIG. **3** schematically shows, in an exploded illustration, a first embodiment of a first modular fastening arrangement **50**.

The first modular fastening arrangement **50** comprises, as lock module **51**, a frame **50**.**1**, on which four lock units **511** are arranged and held. The frame **50**.**1** is, for example, a rectangular carrier or an H-shaped carrier bearing a respective lock unit **511** at its corners or at its free ends. Furthermore, the frame **50**.**1** serves for fastening the seat **1**. For this purpose, four fastening elements **50**.**3** are provided on the frame **50**.**1** and can be used to connect the seat **1** fixedly to the fastening arrangement **50**.

The lock unlocking module **52** comprises at least one first carrier **521**, a lock actuating unit **522**, which is designed as an integral actuating unit **541**, and a lock unlocking mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a longitudinal rod or longitudinal tube. The lock actuating unit **522** is designed as a lever arm. The lock actuating unit **522** comprises a gripping element **522**.**3**. The lock unlocking mechanism **523** is designed, for example, as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms of movement, to the lock units **511** in the assembled state. The lock unlocking mechanism **523** has, for example, two cable pulls **523**.**1** and **523**.**2** which are both designed as double cable pulls.

The rail unlocking module **53** comprises at least one second carrier **531**, a rail actuating unit **532**, which is designed as an integral actuating unit **541**, and a rail unlocking mechanism **533**. The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism or lever mechanism, in particular a cable pull mechanism or lever arm mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail.

Depending on the design of the rail locking means, one locking element **4** can be provided per pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**.

Furthermore, for coupling the lock unlocking module **52** to the lock units **511**, the first fastening arrangement **50** can comprise one or two lock adapters **50**.**2** which can be arranged on the frame **50**.**1**.

Furthermore, the respective lock adapter **50**.**2** can comprise guide elements or sliding elements, in particular rolling elements **6**. In the assembled state of the modular fastening arrangement **5** or **50** and in the fastening arrangements **500** and **5000** described below, the guide elements or sliding elements serve for guiding the seat **1** on or in the rail arrangement **2**, in particular on guide surfaces or rolling surfaces of the lower rails **32** of the pairs of rails **3**. The seat **1** is arranged movably in a sliding manner on the lower rail **31** in the unlocked state of the rail unlocking module **53** using the rolling elements **6** via the respective fastening arrangement **5**, **50**, **500**, or **5000**, which is mounted and arrested on the upper rails **31** by the lock module **51**.

In the exemplary embodiment which is shown, the rolling elements **6** are arranged in a sliding manner on sliding surfaces or guide surfaces of the lower rails **32** (also see in the detail in FIG. **11**). Alternatively, the rolling elements **6** can also be arranged in a sliding manner within the pairs of rails **3**, in a cavity formed between upper rail **31** and lower rail **32**.

FIGS. **4** and **5** schematically show, in an exploded illustration and in a perspective view, a second embodiment of a modular fastening arrangement **500** (called second fastening arrangement **500** below).

Said second fastening arrangement **500** can be preassembled. The second fastening arrangement **500** can be designed as a preassembled docking mechanism **7**.

The carrier or frame **50**.**1** and the lock adapter **50**.**2** are omitted in this embodiment. Said second fastening arrangement **500** therefore comprises fewer parts and modules. In these FIGS. **4** and **5**, the rail arrangement **2** is not illustrated for the sake of clarity.

The second modular fastening arrangement **500** comprises, as lock module **51**, four lock units **511** with integrated carrier elements **512**. The respective lock unit **511** comprises a carrier plate **511**.**12**. The integrated carrier elements **512** serve for fastening the seat **1** in a manner not illustrated specifically. The carrier elements **512** are designed, for example, as fastening bolts or fastening pins which are connected to the seat **1** and to the carrier plate **511**.**12** in the assembled state.

In the assembled state with the rail arrangement **2**, the second fastening arrangement **500** is fixed in a form-fitting and/or force-fitting manner at the rail ends **33** of the upper rails **31** by a respective lock unit **511**.

The four lock units **511** each comprise a blocking element **513** for releasable locking in the rail arrangement **2**. The respective blocking element **513** is designed, for example, as a blocking hook. The blocking hooks are used to releasably fix and fasten the second fastening arrangement **500** on the rail arrangement **2**.

Furthermore, each lock unit **511** comprises an unlocking unit **514** for coupling the lock unit **511** to the actuating module **54**, in particular to one of the actuating units **541**. The unlocking unit **514** is designed as a joint mechanism and/or lever mechanism.

The respective unlocking unit **514** can be used to unlock the associated lock unit **511**, in particular to unlock all the lock units **511** synchronously, such that the lock module **51** can be opened and the second fastening arrangement **500** extracted or removed from the seat **1**.

The unlocking unit **514** is coupled in terms of movement to the rail unlocking module **53** such that, when the rail unlocking module **53** is actuated in order to unlock the locking element **4** of the rail arrangement **2**, the unlocking units **514** are actuated synchronously in order to unlock the lock units **511** of the lock module **51**. As a result, by actuation of the rail unlocking module **51**, for example by an associated rail actuating unit **532**, the locking elements **4** of the rail arrangement **2** and the lock units **511** of the lock module **51** can be unlocked synchronously, and therefore the fastening arrangement **500**, and thus optionally the seat **1**, is released from the rail arrangement **2** and can be extracted.

Two lock units **511**, in particular a front lock unit **511** and a rear lock unit **511**, as seen in the longitudinal direction X of the rail arrangement **2**, are provided per pair of rails **3**. The blocking elements **513** are mounted pivotably on the lock units **511**. The blocking elements **513** are designed as blocking hooks **517**.

The two blocking elements **513** associated with a pair of rails **3** are mounted on the associated lock units **511** in such a manner that the blocking hooks **517** thereof are directed towards one another. In particular, when the unlocking unit **514** is actuated in order to unlock and open the lock units **511**, the two blocking elements **513** of the respective pair of rails **3** are pivoted away from one another. The blocking hooks **517** thereby pivot out of their arresting engagement at the rail ends **33**. The lock module **51** is therefore opened and the second fastening arrangement **500** can be released and extracted from the form-fitting and/or force-fitting connection to the rail arrangement **2**.

The lock unlocking module **52** comprises at least one first carrier **521**, a lock actuating unit **522**, which is designed as an integral actuating unit **541**, and a lock unlocking mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a transverse rod or a transverse tube.

The lock actuating unit **522** is designed as a lever arm mechanism. The lever arm mechanism comprises two lever arms **522.1** and **522.2**, wherein one of the lever arms **522.1** is provided with a gripping element **522.3**. The two lever arms **522.1** and **522.2** are arranged on the carrier **521** for rotation therewith and are coupled in terms of movement via the first carrier **521** such that, when the lever arm **522.1** is actuated, the gripless lever arm **522.2** is entrained by the gripping element **522.3** as a result of the rotation of the first carrier **521**. The lever arms **522.1** and **522.2** are arranged perpendicularly to the first carrier **521**.

The lock unlocking mechanism **523** is designed as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms of movement, to the lock units **511** in the assembled state. The pull mechanism has two cable pulls **523.1** and **523.2** which are both designed as double cable pulls. The two cable pulls **523.1** and **523.2** are respectively coupled at one end to one of the lever arms **522.1** and **522.2** and are coupled at the other end to two associated lock units **511**. By actuation of the lever arm **522.1** and by the other lever arm **522.2** being entrained, the two cable pulls **523.1** and **523.2** are actuated, thus unlocking the associated lock units **511**, in particular the blocking elements **513**. The lock unlocking mechanism **523** is designed as a separate mechanism for unlocking the lock module **51**, in particular for synchronously unlocking all of the lock units **511**.

The rail unlocking module **53** comprises at least one second carrier **531**, a rail actuating unit **532**, which is designed as an integral actuating unit **541**, and a rail unlocking mechanism **533**.

The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism, articulated mechanism and/or bar mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail, and/or to the lock units **511**. In order to actuate the rail unlocking mechanism **533**, the associated integral actuating unit **541** comprises a gripping unit **534**.

Depending on the design of the rail locking means, one locking element **4** can be provided per pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**. Accordingly, the rail unlocking mechanism **533** can comprise one or two rail unlocking units **533.1**.

In order to actuate the rail unlocking mechanism **533**, the rail unlocking module **53** comprises one rail unlocking unit **533.1** per pair of rails **3**. The respective rail unlocking unit **533.1** comprises two lever arms **531.1** which emerge perpendicularly from the carrier **531** and on each of which a free end of a rail unlocking arm **533.2** is formed which actuates the locking element **4** of the respective pair of rails **3** for unlocking or locking purposes.

During the removal or final installation and fastening of the seat **1** from or on the rail arrangement **2**, in said second fastening arrangement **500** the lock unlocking module **52** and the rail unlocking module **53** are coupled to each other in terms of movement. For this purpose, two of the unlocking units **514** are each coupled in terms of movement to one of the rail unlocking units **533.1** of the rail unlocking mechanism **533**.

The unlocking units **514** are designed, for example, as lever arms **514.1** or articulated arms. The rail unlocking units **533.1** are designed as vertically movable pins **533.3** or bolts. Two lever arms **514.1** of the two unlocking units **514** for unlocking the locking element **4** of one of the pairs of rails **3** are coupled to, in particular held pivotably on, the pin **533.3** of one of the rail unlocking units **533.1**.

Such an arrangement permits the synchronous actuation of the rail unlocking module **53**, of the lock module **51**, in particular all of the lock units **511**, and of the lock unlocking module **52**, in particular all of the unlocking units **514**, such that the fastening arrangement **500** can be removed from or mounted on the rail arrangement **2**.

In order to adjust the blocking element **513** from the locking position into the unlocking position, the respective lock unit **511** comprises a control element **511.2**. The control element **511.2** has a guide slot **511.21** (illustrated in FIGS. **5**, **6** and **8**) into which a guide pin **511.8** (illustrated in FIGS. **5** and **6**), which is arranged on the blocking element **513**, is guided.

The construction and the components of the respective lock unit **511** are described in more detail below with reference to FIG. **8**.

FIGS. **6** and **7** schematically show, in an exploded illustration and in a perspective view, a third embodiment of a modular fastening arrangement **5000** (called third fastening arrangement **5000** below). The carrier or frame **50.1** and the lock adapter **50.2** of the first fastening arrangement **50** are omitted in this embodiment. Said third fastening arrangement **5000** therefore comprises fewer parts and modules. In these FIGS. **6** and **7**, the rail arrangement **2** is not illustrated for the sake of clarity.

The third modular fastening arrangement **5000** comprises, as lock module **51**, four lock units **511** with integrated carrier elements **512**. The integrated carrier elements **512** serve for fastening the seat **1** in a manner not illustrated specifically. The third modular fastening arrangement **5000** differs in the type of lock unlocking means which comprises a cable pull mechanism instead of an articulated mechanism as the unlocking unit **514**.

The four lock units **511** each comprise, analogously to the second modular fastening arrangement **500**, a blocking element **513** for the releasable locking on or in the rail arrangement **2**. The respective blocking element **513** is designed, for example, as a blocking hook **517**.

Furthermore, a pair of lock units **511** is in each case coupled in terms of movement to one another by an associated unlocking unit **514**, for example unlocking lever or unlocking arm. For example, the unlocking unit **514** is designed as a deflecting pulley **515** which is actuable by the lock actuating unit **522**. Instead of a lever arm **522.1**, **522.2**, the lock actuating unit **522** comprises a lock actuating cable **522.4**. A gripping element **522.3**, in particular a cable loop, is arranged at the free end of the lock actuating cable **522.4**.

The lock unlocking module **52** comprises, as first carrier **521**, a transmission rod, the lock actuating unit **522**, which is designed as an integral actuating unit **541**, and the lock unlocking mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a transverse rod or a transverse tube.

The lock actuating unit **522** is designed as a cable pull mechanism. The cable pull mechanism comprises the lock actuating cable **522.4** which is provided with the gripping element **522.3**. The lock actuating cable **522.4** is arranged via the deflecting pulley **515** on the first carrier **521** for rotation therewith, such that, when the lock actuating cable **522.4** is actuated, the first carrier **521** is rotated at the same time by the gripping element **522.3** in order to actuate the unlocking unit **514**, in particular the deflecting pulley **515**, which, in turn, is coupled, in particular coupled in terms of movement, to the lock unlocking mechanism **523**.

The lock unlocking mechanism **523** is designed as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms of movement, to the lock units **511** in the assembled state. The cable pull mechanism has two cable pulls **523.1** and **523.2** which are both designed as a lock unlocking cable. The two cable pulls **523.1** and **523.2** are respectively coupled at one end to one of the front lock units **511** and are coupled at the other end to one of the rear lock units **511**. For this purpose, each lock unit **511** has an associated winding unit **524**.

By actuation of the lock actuating cable **522.4** and driving the deflecting pulley **515**, the two cable pulls **523.1** and **523.2** are actuated, as a result of which the associated lock units **511** are unlocked or locked by winding said cable pulls onto or unwinding them from the winding units **524**.

The rail unlocking module **53** comprises at least the second carrier **531**, the rail actuating unit **532**, which is designed as an integral actuating unit **541**, and the rail unlocking mechanism **533** (not illustrated specifically in FIGS. **6** and **7**, and constructed analogously as shown in FIGS. **4** and **5** and described with regard thereto).

The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism, articulated mechanism and/or bar mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail, and/or to the lock units **511**. In order to actuate the rail unlocking mechanism **533**, the associated integral actuating unit **541** comprises the gripping unit **534**.

Depending on the design of the rail locking means, one locking element **4** can be provided per pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**. Accordingly, the rail unlocking mechanism **533** can comprise one or two rail unlocking units **533.1**.

During the removal or final installation and fastening of the seat **1** from or on the rail arrangement **2**, in said third fastening arrangement **5000** the lock unlocking module **52** and the rail unlocking module **53** are coupled in terms of movement to one another. For this purpose, the respective deflecting pulley **515** is coupled in terms of movement to the rail unlocking mechanism **533**, in particular is coupled in terms of movement to an associated rail unlocking arm **533.2**.

FIG. **8** schematically shows, in an exploded illustration, an exemplary embodiment of a lock unit **511**. All of the lock units **511** of the lock module **51** are constructed analogously.

The respective lock unit **511** comprises at least:

- one first spring element **511.1** (also called "spring control element") for a control element **511.2** (also called "control element")
- one second spring element **511.3** (also called "spring crash cam"),
- two carrier elements **512** (also called "bolt cam")
- one compensating spring **511.4** (also called "spring tolerance cam") for compensating for tolerances between the carrier elements **512**,
- one blocking hook holder **511.5** (also called "bolt hook") for the blocking hook **517**,
- one holding pin **511.6** (also called "pin control element") for the control element **511.2**,
- the unlocking unit **514** (also called "link track rod"),
- one coupling pin **511.7** (also called "pin cam link"),
- the blocking element **513** with the blocking hook **217** (also called "crash hook"),
- two guide pins **511.8** (also called "pin ISO 8734 6×20"),
- one compensating element **511.9** (also called "tolerance cam"),
- one crash cam element **511.10** (also called "crash cam"),
- one reinforcing element **511.11** (also called "reinforcement plate"), optionally for reinforcing the carrier plate **511.12**,
- one carrier plate **511.12** (also called "side plate"),
- one triggering plate **511.13** (also called "release plate"),
- one third spring element **511.14** for a triggering arm (also called "spring release lever"), and
- one bearing bushing **511.15** for the third spring element **511.14** (also called "bush spring").

FIG. **9** shows the lock unit **511** preassembled in the assembled state as a preassembly unit.

The triggering plate **511.13** comprises an entraining pin **511.16** as a guide pin **511.8**. Furthermore, the triggering plate **511.13** comprises a receptacle **511.17**, in particular a slot, for receiving and holding one end of the associated cable pull **523.1**. The third spring element **511.14** is designed as a restoring spring.

The respective lock module **51** can be actuated differently:

- actuation by the lock unlocking mechanism **523** or
- actuation via the unlocking unit **514** by the rail unlocking module **53** and thus synchronous actuation of the rail unlocking mechanism **533** for the locking elements **4** of the rail arrangement **2** and of the unlocking units **514** for the lock units **511**.

When the cable pull **523.1** is actuated to unlock the lock unit **511**, in particular to pivot the blocking hook **517** and therefore to release the blocking hook **517** from the upper rail **31**, the triggering plate **511.13** is entrained and moved, in particular rotated, by the entraining pin **511.16**. As a result, the third spring element **511.14** is tensioned and the blocking element **513** and therefore the blocking hook **517** are pivoted. The blocking hook **517** is positioned into an unlocking position and thus released from the upper rail **31**.

If the actuation is ended, the triggering plate **511.13**, the blocking element **513** and the blocking hook **517** are reset by the third spring element **511.14** into their starting position which corresponds to a locking position.

During the actuation of the unlocking unit **514** by the rail unlocking module **53**, in particular the rail unlocking unit **533.1**, the crash cam element **511.10** is forcibly guided, in particular rotated by the guide pin **511.8** about an axis of rotation **511.18**. The guide pin **511.8** arranged on the cam element **511.10** is guided in a guide slot **511.19**. For this purpose, the crash cam element **511.10** is connected in an articulated manner to the unlocking unit **514** at the point of articulation **511**.**20**. The crash cam element **511**.**10** entrains the blocking element **513** by the coupling via the compensating element **511**.**9** and the control element **511**.**2**. The blocking element **513** is moved from the locking position into the unlocking position. For this purpose, the control element **511**.**2** has the guide slot **511**.**21** (illustrated in FIGS. **5**, **6** and **8**) into which the guide pin **511**.**8** (illustrated in FIGS. **5** and **6**), which is arranged on the blocking element **513**, is guided when the blocking element **513** is moved from the locking position into the unlocking position.

FIG. **10** shows a schematic illustration of a releasable seat **1** on the rail arrangement **2**. In order to install the seat **1**, the latter is to be arranged on the rail arrangement **2**, in a region in which the lower rail **32** is free of the upper rail **31**.

FIG. **11** shows a schematic illustration of the seat **1** pre-positioned on the rail arrangement **2**. The seat **1** is pre-positioned here by rolling elements **6** on its lower side on the lower rail **32**. The lock module **51** is open. The seat **1** is guided on sliding surfaces of the lower rail **32** (or alternatively of the upper rail **31**, not illustrated) by the rolling elements **6** and pre-positioned in such a manner that the lock units **511** are positioned above the rail ends **33** of the upper rail **31**.

FIGS. **12** to **20**F show schematic illustrations of various embodiments and the movement sequences thereof during the unlocking or locking of one of the fastening arrangements **5**, **50**, **500**, **5000** from or on the rail arrangement **2**.

FIG. **12** shows, as actuating module **54**, a lever arrangement which, in order to release the fastening arrangement **5**, **50** or **500**, is actuable downwards in the direction of the rail arrangement **2**, according to arrow P1, in order to allow the blocking elements **513** of the lock units **511** to drop into the lower rails **32** of the rail arrangement **2**, in particular into a receiving slot of the U-shaped lower rails **32**. For the description of the fastening arrangements **5**, **50** and **500**, reference is made to FIGS. **1** to **5** and **8** to **11**.

In FIG. **12**, the fastening arrangement **5**, **50** or **500** is shown in the unlocked or open state. The lever arms **514**.**1** and therefore the blocking elements **513** are raised here as a result of the upwardly actuated actuating module **54**, in particular the actuating unit **541**. In this state, the seat **1** with the fastening arrangement **5**, **50** or **500** can be pre-positioned on the lower rail **32** by the rolling elements **6**.

The seat **1** comprises two lateral seat carriers **1**.**1**. Each pair of rails **3** is assigned a seat carrier **1**.**1**.

The lock units **511** with the blocking elements **513** are each arranged here on an outer side of the seat carriers **1**.**1**. The rolling elements **6** are arranged on an inner side of the seat carriers **1**.**1**. The actuating module **54** is arranged between the two seat carriers **1**.**1** and connected via passage openings in the seat carriers **1**.**1** to the outer lock module **51**. Components of the lock unlocking module **52** and/or of the rail unlocking module **53** are partially or completely arranged on the inner side and/or the outer side of the seat carriers **1**.**1**.

When the actuating unit **541** is actuated downwards according to arrow P1, the blocking elements **513**, in particular the blocking hooks **517**, then engage in an arresting manner on the rail ends **33** of the upper rail **31**. In particular, the blocking elements **513** of the lock units **511** drop into the respective receiving slot of the U-shaped lower rails **32** and reach under the rail ends **33** of the upper rail **31** such that the seat **1** is fastened in a vertically secured manner to the rail arrangement **2**.

FIG. **13** shows, as actuating module **54**, the previously described cable pull mechanism which, in order to release the third fastening arrangement **5000**, is actuable upwards away from the rail arrangement **2**, according to arrow P**2**, in order to release the blocking elements **513** of the lock units **511** from the rail arrangement **2**.

In order to arrest the blocking hooks **517** on the respective upper rail **31**, the respective upper rail **31** comprises a hook receptacle **31**.**1** at each of its rail ends **33**. The upper rail **31** is arranged movably in the lower rail **32** by further rolling elements **6**, in particular in each case two rolling elements **6** in the region of each rail end **33**. In particular, the hook receptacles **31**.**1** form the respective rail end **33**. The rolling elements **6** are arranged in the upper rail **31** between the two hook receptacles **31**.**1**, in particular substantially adjacent to them, and roll on a base surface of the lower rail **32**.

For the description of the fastening arrangements **5** and **5000**, reference is made to FIGS. **1**, **6** and **7**, and **10** to **11**.

FIG. **14**A shows a side view of the fastening arrangement **5**, **50** or **500** with the docking mechanism **7** in order to mount or to remove the respective fastening arrangement **5**, **50** or **500** on or from the rail arrangement **2**. The side view means that only half of the fastening arrangement **5**, **50** or **500** is shown with the two lock units **511** and the lock unlocking mechanism **523** with one of the cable pulls **523**.**1** for one of the pairs of rails **3**. The other part of the docking mechanism **7** for the other pair of rails **3** is constructed analogously, as previously described. The actuating module **54** is configured in such a manner that it can actuate the two parts synchronously, as has been previously described in more detail with reference to FIGS. **1** to **5** and **8** to **13**.

FIG. **14**A shows, in the side view, that side of the carrier plate **511**.**12** of the respective lock unit **511** which forms an outer side in the mounted state on the seat carriers **1**.**1**.

FIG. **14**B shows an unlocking mechanism **8** which is provided on the opposite side of the carrier plate **511**.**12** of the respective lock unit **511** in order to arrest the respective fastening arrangement **5**, **50** or **500** on the rail arrangement **2** for the fastening of the seat **1** or to release same therefrom and to unlock the rail locking means, in particular the locking elements **4** of the pairs of rails **3**, synchronously.

The unlocking mechanism **8** is designed as a lever mechanism and/or articulated mechanism for unlocking the rails in combination with a cable mechanism for unlocking the lock.

FIG. **15** shows an alternative unlocking mechanism **80** which is designed as a cable pull mechanism and corresponds to the previously described third fastening arrangement **5000**. FIG. **15** shows the fastening arrangement **5000** which is arranged between the seat **1** and the rail arrangement **2** and releasably locks the seat **1** to the rail arrangement **2** using the lock units **511**.

FIGS. **16**A and **16**B show the blocking elements **513** of two lock units **511** on one of the pairs of rails **3** being unlocked by a lever mechanism corresponding to the unlocking mechanism **8**.

By pulling the rail unlocking arm **533**.**2** upwards in accordance with arrow P3, the rail unlocking mechanism **533** is actuated, in particular the second carrier **531**, which is designed as an unlocking rod, is rotated such that the unlocking units **514** are actuated in accordance with arrow P4 and the locking element **4** is unlocked, and thus the rails **31**, **32** are released. The release of the rails **31**, **32** in the "vis-à-vis" position leads to increased complexity since the unlocking rod has to be rotated in the same direction both on the inner side and on the outer side of the carrier plate **511**.**12**, with the pulling direction on one side being switched over in the opposite direction.

FIGS. **17**A and **17**B show the blocking elements **513** of two lock units **511** on one of the pairs of rails **3** being unlocked by a cable pull mechanism corresponding to the unlocking mechanism **80** for the third fastening arrangement **5000**. The rail actuating unit **532** is pivoted or pulled upwards in accordance with arrow P**5**. As a result, the unlocking unit **514** is moved, in particular pivoted, downwards, in accordance with arrow P**6**, in order to unlock the rail locking means, in particular the locking element **4**.

The direct unlocking of the rail arrangement **2** is undertaken by pressing down the locking element **4**, for example an unlocking plate, vertically in the upper rail **31** (also called rail runner) and reduces the complexity and possible problems when latching the unlocking unit **514**, for example an unlocking lever, unlocking arm or unlocking pin, into an unlocking bushing of the upper rail **31** (rail runner) during the docking operation. The triggering mechanism or unlocking mechanism according to circle K functions both in the "forwardly directed" position and in the "vis-à-vis" position.

FIG. **18**A shows the seat **1** which rolls along the lower rail **32** in the direction of the upper rail **31** in accordance with arrow P**7**. The blocking elements **513** are guided in slot-shaped openings in the U-shaped lower rail **32**.

FIG. **18**B shows the lifting of the front blocking element **513** of the front lock unit **511** out of the lower rail **32**, wherein, during the further movement of the seat **1** in the direction of the preassembled upper rail **31**, the blocking element **513** is pushed onto the upper rail **31** and moved along the latter.

FIG. **18**C shows the state of the fastening arrangement **5** (**50**, **500** or **5000**) after the seat **1** has been moved forward to such an extent that the upper rail **31** strikes against the rear blocking element **513**, wherein, owing to restoring elements, the front blocking element **513** latches again automatically into the upper rail **31** in accordance with arrow P**8** and therefore the two lock units **511** are fully latched into place. In this position, the rail unlocking unit **533**.**1**, in particular a track rod, drops into the upper rail **32** (also called running rail or track slider) such that an arresting connection in accordance with arrow P**9** is produced and the seat **1** is arrested on the rail arrangement **2** by the fastening arrangement **5**, **50** or **500** and the locking element **4** that has been latched into place.

FIGS. **18**D and **18**E show an actuation of the lock actuating unit **522** in the upwards direction in accordance with arrow P**10** firstly for unlocking the rails via the actuated unlocking units **514**, in particular unlocking levers. The unlocking units **514**, in particular unlocking levers, are connected via cable pulls **523**.**1**, in particular via Bowden cables, to the locking element/locking elements **4**, in particular locking plates, and lift them in accordance with arrows P**11** and P**12**. In the process, the triggering plates **511**.**13** of the lock units **511** are rotated and lift the control elements **511**.**2** and, with the latter, the blocking elements **513**, upwards until the blocking elements **513** are released from the locking region on the upper rail **31** and are therefore unlocked. In parallel therewith, the locking of the rails is released by the locking element **4** being moved, in particular lifted, into a release position and out of the upper rail **31** and the lower rail **32**.

FIG. **18**F shows the further actuation of the lock actuating unit **522** into a further raised position, wherein the triggering plates **511**.**13** are rotated further and the control elements **511**.**2** and the blocking elements **513** are raised further until the blocking elements **513** are arranged above the respective pair of rails **3** in accordance with arrow P**13**. In this release position or unlocking position of the fastening arrangement **5**, **50**, **500** or **5000**, the lock actuating unit **522** is held secured in the associated raised position by the third spring element **511**.**14** (also called softlock spring). The seat **1** can now be rolled on the rail arrangement **2** or removed from the latter.

FIGS. **19**A to **19**C show a cable mechanism as actuating module **54**.

FIG. **19**A shows an actuation of the lock actuating cable **522**.**4** upwards in accordance with arrow P**14**. The deflecting pulley **515** (also called central pulley) is thereby moved in the anticlockwise direction, in particular rotated in accordance with arrow P**15**.

FIG. **19**B shows that, as a result of the rotational movement of the deflecting pulley **515**, the cable pulls **523**.**1** and **523**.**2** are actuated and raise a cam disc **516**, in particular pivot the latter upwards, in accordance with arrow P**16**.

FIG. **19**C shows that, upon further rotation of the deflecting pulley **515**, the blocking elements **513** and the control elements **511**.**2** are raised in accordance with arrow P**17**. In order to hold the fastening arrangement **5** or **5000** in this release position or unlocked position, said fastening arrangement comprises a cable arresting means **9** (also called pull-pull lock) which arrests the cable pulls **523**.**1** and the lock actuating cable **522**.**4**. The cable arresting means **9** is designed as a rotatable arresting hook or blocking lug **9**.**1**.

FIGS. **20**A to **20**F show the sequence of movement of the cable arresting means **9**.

FIG. **20**A shows the deflecting pulley **515** which rotates during the unlocking operation in accordance with arrow P**18** and, in the process, entrains the cable arresting means **9**, in accordance with arrow P**19**, by an entraining pin **10**, which also serves as arresting pin **9**.**2**.

FIG. **20**B shows the cable arresting means **9** in the arresting position in which it arrests the cable pulls **523**.**1** and the lock actuating cable **522**.**4** and the fastening arrangement **5**, **5000** and therefore the blocking elements **513** are placed in the fully released position (release position). The cable arresting means **9** springs back until its blocking lug **9**.**1** strikes against an arresting pin **9**.**2**, in accordance with arrow P**20**.

FIG. **20**C shows the cable arresting means **9** in a first secured position GS**1** when the lock actuating cable **522**.**4** is released again. The seat **1** can be moved, in particular rolled along the upper rail **31**, or can be removed from the rail arrangement **2**.

FIG. **20**D shows the cable arresting means **9** in a second secured position GS**2** in accordance with arrow P**21**, into which it is brought when the lock actuating cable **522**.**4** is actuated a further time. For this purpose, the blocking lug **9**.**1** has a blocking contour **9**.**3** which has correspondingly shaped blocking receptacles for the first and second secured position GS**1**, GS**2** for the arresting pin **9**.**2**.

FIG. **20**E shows the cable arresting means **9** when the lock actuating cable **522**.**4** is released. In the process, the arresting pin **9**.**2** moves in the Y axis direction and below the cable arresting means **9** along the surface side thereof, in accordance with arrow P**22**.

FIG. **20**F shows the automatic, in particular spring-supported, restoring of the lock actuating cable **522**.**4** and of the deflecting pulley **515** into their starting position or locking position, in which the blocking elements **513** move downwards and therefore drop into the rail arrangement **2** when the seat **1** is arranged together with the associated fastening arrangement **5**, **50**, **500** or **5000**. The cable arresting means **9** is released in the process. The arresting pin **9**.**2** is disengaged.

LIST OF REFERENCE DESIGNATIONS

1. Seat
1.1 Seat carrier

2 Rail arrangement
3 Pair of rails
31 Upper rail
31.1 Hook receptacle
32 Lower rail
33 Rail end
4 Locking element
5 Fastening arrangement
51 Lock module
511 Lock unit
511.1 First spring element
511.2 Control element
511.3 Second spring element
511.4 Compensating spring
511.5 Blocking hook holder
511.6 Holding pin
511.7 Coupling pin
511.8 Guide pin
511.9 Compensating element
511.10 Crash cam element
511.11 Reinforcing element
511.12 Carrier plate
511.13 Triggering plate
511.14 Third spring element
511.15 Bearing bushing
511.16 Entraining pin
511.17 Receptacle
511.18 Axis of rotation
511.19 Guide slot
511.20 Point of articulation
511.21 Guide slot
512 Integrated carrier element
513 Blocking element
514 Unlocking unit
514.1 Lever arm
515 Deflecting pulley
516 Cam disc
517 Blocking hook
52 Lock unlocking module
521 First carrier
522 Lock actuating unit
522.1, **522.2** Lever arm
522.3 Gripping element
522.4 Lock actuating cable
523 Lock unlocking mechanism
523.1, **523.2** Cable pull
524 Winding unit
523 Rail unlocking module
531 Second carrier
531.1 Lever arm
532 Rail actuating unit
533 Rail unlocking mechanism
533.1 Rail unlocking unit
533.2 Rail unlocking arm
533.3 Pin
534 Gripping unit
54 Actuating module
541 Actuating unit
50 First modular fastening arrangement
50.1 Frame
50.2 Lock adapter
50.3 Fastening element
500 Second modular fastening arrangement
5000 Third modular fastening arrangement
6 Rolling element
7 Docking mechanism
8, **80** Unlocking mechanism
9 Cable arresting means
9.1 Blocking lug
9.2 Arresting pin
9.3 Blocking contour
10 Entraining pin
GS**1** First secured position
GS**2** Second secured position
K Circle
P**1** to P**22** Arrow
X Longitudinal direction
Y Transverse direction
Z Vertical direction

The invention claimed is:

1. A fastening arrangement of a seat on a rail arrangement, wherein the fastening arrangement is of modular construction and comprises:

one lock module for releasably arresting the fastening arrangement on the rail arrangement,

wherein the lock module is of multi-part design and comprises at least two lock units,

wherein each of the lock units are releasably fixed to both the seat and the rail arrangement, and

one unlocking module for unlocking the lock module,

wherein at least two rolling elements are rotatably mounted to the lock module so that the fastening arrangement is rolled away on the rail arrangement when the lock module is unlocked.

2. The fastening arrangement according to claim **1**, wherein the lock units are arranged on a frame or on a carrier plate.

3. The fastening arrangement according to claim **1**, wherein the unlocking module is configured to unlock the lock units synchronously from the rail arrangement.

4. The fastening arrangement according to claim **1**, wherein an actuating module actuates the unlocking module of the fastening arrangement and/or a rail unlocking module.

5. The fastening arrangement according to claim **4**, wherein the actuating module is coupled to the lock module via the unlocking module.

6. The fastening arrangement according to claim **4**, wherein the actuating module is coupled to a locking element for the rail arrangement via the unlocking module.

7. A seat with a rail arrangement and a fastening arrangement according to claim **1** for releasably fastening the seat on the rail arrangement.

8. The seat according to claim **7**, wherein the fastening arrangement is designed as a preassemble docking mechanism.

9. The seat according to claim **7** wherein the rail arrangement comprises two pairs of rails each having an upper rail and a lower rail, and the fastening arrangement is releasably arrestable at one rail end of the upper rail by a respective lock unit.

* * * * *